

Health Impacts of Food Away From Home

Victoria Milani

July 30, 2026

Abstract

This study examines the impact of Food Away from Home (FAFH) consumption on health for males and females, using indices of metabolic, cardiovascular, and body-shape health, adjusted self-reported health, and diet quality measured by the Healthy Eating Index (HEI). Using data from the National Health and Nutrition Examination Survey (NHANES), the analysis employs a bunching design at zero FAFH calories, following Caetano et al. (2024a), with the latent demand for FAFH recovered through a control-function representation. The physiological effects of FAFH concentrate among females: an additional 1,000 daily FAFH calories raises the body-shape index by 0.17 standard deviations, the metabolic index by 0.07, and the cardiovascular index by 0.05, where higher values indicate worse health. Males show no significant response on any physiological index. Diet quality is the margin both genders share: an additional 1,000 FAFH calories lowers the HEI by 1.9 points for females and 1.3 for males, and the bunching design attributes roughly half of the naive diet-quality gap to selection rather than to the food itself.

1 Introduction

Americans have moved a third of their eating out of the kitchen. Food away from home (FAFH), that is everything obtained from restaurants, fast-food outlets, cafeterias, vending machines, and other sources outside the household’s own food supply, provided 33% of Americans’ caloric intake by 2018, nearly double its share in the late 1970s (Lin & Guthrie, 2012; Saksena et al., 2018). Spending shows the same shift: in 2018, 44 cents of every food dollar went to FAFH (Bureau of Labor Statistics, 2020), and the average American ate away from home five times a week (Saksena et al., 2018).

The food itself is different. FAFH carries a third more sodium per calorie than food prepared at home, 1,820 milligrams per 1,000 calories against 1,369, and comes in larger, more calorie-dense portions (Binkley, 2008; Lin & Guthrie, 2012; Prentice & Jebb, 2003). Adults who eat out at least weekly take in roughly 80 more calories per day than those who rarely do, along with less fiber and about 0.3 fewer daily portions of fruits and vegetables (Goffe et al., 2017; Seguin et al., 2016). Eating out also tracks poor health: FAFH consumption is associated with higher BMI and obesity risk (Binkley et al., 2000; Bowman & Vinyard, 2004; Pereira et al., 2005), insulin resistance (Pereira et al., 2005), and worse biomarkers for chronic disease (Kant et al., 2015).

None of these associations says what eating out does to health, because none of them compares like with like. People who eat out frequently differ from people who do not, in age, income, work

schedules, and household structure, and in habits and preferences that no dataset records. A comparison of the two groups mixes the effect of restaurant food with the effect of being the kind of person who chooses it, and there is no reason to expect either component to be small, or the two to share a sign. This paper separates them.

I estimate the causal effect of FAFH calories on adult health using the bunching design of Caetano et al. (2024a), implemented as a control function. The data are the 2005–2018 waves of the National Health and Nutrition Examination Survey (NHANES), covering roughly 30,000 adults aged 22 to 80 who reported a typical eating day. The treatment is the number of FAFH calories reported in a 24-hour dietary recall. There are five outcomes: three standardized indexes of metabolic, cardiovascular, and body-shape health built from examination and laboratory measures; an adjusted measure of self-reported health; and the Healthy Eating Index (HEI), the USDA’s summary score of diet quality. Females and males differ in metabolism and body composition, so all results are estimated separately by gender.

FAFH harms health, and it harms females on more margins than males. The indexes are standardized so that higher values mean worse health. For females, an additional 1,000 FAFH calories per day raises the body-shape index by 0.169 standard deviations, the metabolic index by 0.074, and the cardiovascular index by 0.054. For males, no physiological index responds detectably; the male harm runs through diet quality alone, as the next paragraph shows. The effect on self-reported health is a null for both genders, so for females the damage shows up in biomarkers but not in how people say they feel.

To fix magnitudes, a 1,000-calorie increment is a comparison the data actually contain rather than an extrapolation: males aged 22–44 average 1,026 FAFH calories per day and females 636, so the estimates compare a typical young consumer with someone who eats out lightly. For scale, Hoynes et al. (2016) estimate that full childhood exposure to food stamps improves an adult metabolic-health index by about 0.3 standard deviations; the female responses here run from a fifth to over half that size, from a single dietary margin. Female effects are also stable across age groups; the male age profile is noisier, and Section 5 examines it in detail.

The diet-quality penalty survives the correction, but only half of it. In the raw data, an additional 1,000 FAFH calories predicts an HEI about 4 points lower for females and 2 points lower for males once demographics are held fixed, which reads as confirmation that eating out means eating badly. The bunching design splits that association in two: a causal effect of -1.9 HEI points per 1,000 kcal for females and -1.3 for males, both significant, and selection of comparable size (-1.7 and -0.7). Eating out does degrade the diet at the margin, and the people who eat out most would also eat worse diets regardless. The naive comparison is not wrong about the direction, but it overstates the causal penalty by roughly a factor of two.

The causal reading of these estimates rests on how the design separates the effect of FAFH from selection into FAFH. Consumption cannot be negative, so everyone who would ideally consume less than zero, however strong their distaste for eating out, ends up at the same observed value. About 30% of the sample reports exactly zero FAFH calories. These bunched respondents eat

essentially the same amount of FAFH as the lightest consumers, so the health gap between the two groups cannot be an effect of the food. It measures the selection directly. Two assumptions turn that measurement into an estimate. Linear endogeneity requires the strength of selection to be constant across the range of latent FAFH demand, so the slope measured at the bunching point extends to the rest of the distribution. Tail symmetry pins down how far below zero the bunched group’s latent demand lies, by requiring its lower tail to mirror the observed upper tail of positive consumption. Because the mirrored tail is the empirical one rather than a fitted curve, this is the conservative choice among the distributional assumptions considered, and Section 6 re-estimates everything under the alternatives. Subtracting the implied selection from the observed relationship leaves the causal effect.

The closest causal evidence on FAFH and health comes from two designs: first-differencing that removes time-invariant individual heterogeneity (Mancino et al., 2009), and geographic variation in fast-food exposure (Currie et al., 2010). Both identify from changes in exposure, over time or across space, and both stop at dietary intake or obesity; neither observes the metabolic and cardiovascular margins studied here. The bunching design needs no instrument, no policy discontinuity, and no panel structure, only the mass of non-consumers that every dietary survey exhibits. The framework comes from Caetano et al. (2024a) and has been applied to maternal labor supply by Caetano et al. (2024b); this paper brings it to FAFH and health. Section 2 develops both strands of the literature.

Section 2 reviews the related literature. Section 3 describes the data, the construction of the treatment and outcome variables, and dietary patterns in the sample. Section 4 presents the empirical strategy. Section 5 reports the main results and heterogeneity, Section 6 the robustness checks, and Section 7 concludes.

2 Literature Review

Two literatures meet in this paper: the economics of eating out and its health consequences, and the econometric literature on bunching, with a final word on how the outcomes are measured.

Economists first connected the restaurant sector to health through obesity. Cutler et al. (2003) attribute the rise in American caloric intake to the falling time cost of prepared food, and Chou et al. (2004) link adult obesity to the expansion of restaurants, which together with food prices explains much of the rise in obesity. A large descriptive literature fills in the dietary mechanics: eating out is associated with higher caloric intake, less fiber and fewer fruits and vegetables (Bowman & Vinyard, 2004; Seguin et al., 2016), with weight gain and insulin resistance over long horizons (Pereira et al., 2005), and with adverse chronic-disease biomarker profiles (Kant et al., 2015). The patterns differ by restaurant type (Duffey et al., 2007; Larson et al., 2011) and persist in recent NHANES waves (Nagao-Sato & Reicks, 2022). These are comparisons of people who eat out more with people who eat out less, so they inherit the selection problem described in the introduction.

A smaller literature pursues causal designs, along two routes. The first uses within-person variation: Mancino et al. (2009) difference across each respondent’s two dietary-recall days, so that

stable individual traits drop out, and find that each FAFH meal adds roughly 134 calories and lowers daily diet quality, a result Todd et al. (2010) confirm for adult diet quality with the same source of variation. The second uses geography: Currie et al. (2010) measure the effect of fast-food proximity, finding roughly 5% higher obesity rates among ninth graders within a tenth of a mile of a fast-food restaurant; Alviola et al. (2014) find similar effects at the school level, and Abrahamsson et al. (2023) trace childhood and adolescent exposure to fast-food restaurants through to BMI and cognitive ability. Two limits remain. The geographic designs concern fast food, mostly for children and adolescents, and identify from the consumers whose behavior responds to a nearby outlet. And across both routes, the outcomes stop at intake, diet scores, and body weight; none reaches the physiological margins where diet is supposed to do its damage.

The estimates here cross both limits. The treatment is all FAFH, not only fast food, for adults aged 22 to 80. The outcomes are biomarker-based indexes of metabolic, cardiovascular, and body-shape health alongside diet quality and self-reported health, so the estimates speak to physiology rather than only to weight. And the design separates channels the earlier ones could not: the physiological effects concentrate among females, and roughly half of the diet-quality penalty that looms large in raw comparisons turns out to be selection, with the remainder a real effect of the marginal FAFH calorie.

The identification strategy comes from the bunching literature. Bunching designs originate in public finance, where mass points at kinks and notches of tax schedules reveal behavioral responses to incentives (Kleven, 2016; Saez, 2010). The variant used here differs in where the bunching comes from: not a policy-created kink in the choice set, but the non-negativity constraint on the treatment itself. Caetano et al. (2024a) develop the framework, in which the mass of observations at zero both reveals and corrects the endogeneity bias. The design has since been applied to maternal labor supply (Caetano et al., 2024b), children’s household chores (Caetano et al., 2026), enrichment activities (Caetano et al., 2024c), and video watching (Caetano et al., 2025). This paper applies it to FAFH and health, where the design’s one requirement, an excess mass at zero, comes free with the data: roughly 30% of adults in the sample report no FAFH calories at all.

Finally, the outcome variables follow two established practices. Standardized summary indexes of biomarkers raise power to detect effects that move several markers in the same direction (Evans & Garthwaite, 2014; Hoynes et al., 2016; Kling et al., 2007), a practice with antecedents in epidemiology (Seeman et al., 2008). Biomarkers paired with self-reported health are, in turn, standard outcomes for measuring the health consequences of economic circumstances (Bania & Leete, 2022; Michaud et al., 2016; Schaller & Stevens, 2015). Section 3.2 details the construction.

3 Data

The National Health and Nutrition Examination Survey (NHANES) is a nationally representative survey, released in two-year cycles, that interviews and physically examines approximately 5,000 individuals per year to assess the health and nutritional status of American adults and children.

The interview collects demographic, socioeconomic, dietary, and health-related information. The examination adds medical, dental, and physiological assessments together with laboratory tests. Dietary intake comes from a 24-hour recall that records every food and beverage consumed, along with its source, eating occasion, quantity, nutrient content, and calories. A second recall follows by telephone 3 to 10 days later, to capture day-to-day variation. Participants also state whether each interview day reflects typical, higher-than-usual, or lower-than-usual eating.

The analysis pools the seven NHANES waves from 2005–2006 through 2017–2018 (cycles D through J) and keeps individuals aged 22 to 80, adults past the dietary and metabolic instability of adolescence but young enough that survivorship does not dominate the sample. Pregnant women are excluded, as are respondents missing data on key variables such as caloric intake and those who report the recall day as an atypical eating day. The estimation samples contain between roughly 13,700 and 15,300 respondents per gender, with the count varying across outcomes because not every respondent provides every biomarker (the exact N appears beneath each row of the results tables).

The sample splits into three age groups (Table 1) chosen to track life stages: early working years (22–44), the years surrounding menopause for women (45–64), and retirement age (65 and older).

Table 1: Number of Observations by Groups

		Age Groups		
		22-44	45-64	65+
Gender	Male	7,674	6,415	4,846
	Female	7,697	6,755	4,842

Each cell from this table shows the number of observations for each age bin separated by gender.

Section 3.1 defines the treatment variable, Section 3.2 describes the outcomes, Section 3.3 covers the controls, and Section 3.4 closes with stylized facts about how Americans eat.

3.1 Treatment Variable

The treatment variable is the amount of calories consumed in Food Away From Home (FAFH) on the first 24-hour dietary recall, denoted C_i for individual i . NHANES collects two 24-hour dietary recalls per respondent, the first in person at the Mobile Examination Center and the second by telephone 3 to 10 days later. Following Ricciuto et al. (2021) and the NIH guidance that a single 24-hour recall is sufficient for population-level analysis,¹ the headline analysis uses the first recall only. Participation in the second recall is voluntary, which would introduce selection bias if used as the primary measure, so the second recall and the two-recall average enter only through the treatment-definition robustness in Section 6.4.

¹See “... you can use data from a single administration of a 24HR and you do not need to adjust for within-person random error,” (National Cancer Institute, [n.d.](#)).

The FAH/FAFH classification follows the ERS-USDA convention. FAH includes foods purchased at supermarkets, grocery stores, residential dining facilities, foods grown or caught by the household, and gifts from another person. FAFH includes all remaining sources, with the main contributors being fast-food and full-service restaurants. Table A.1 in Appendix A gives the complete mapping from NHANES food-source codes to this classification.

Each respondent’s recall day decomposes into FAH calories (C_i^{FAH}) and FAFH calories ($C_i^{\text{FAFH}} \equiv C_i$), with the latter used as the treatment throughout. A non-trivial share of respondents report zero FAFH calories on the recall day, and this mass point at $C_i = 0$ is what the bunching design of Section 4 exploits. Figure 10 in Section 4.2 displays the empirical distribution of C_i in the full sample.

3.2 Outcome Variables

The outcomes cover four health domains — metabolic, cardiovascular, body shape, and self-perceived health — plus overall diet quality measured by the Healthy Eating Index (HEI). The physical measures reflect long-run dietary exposure rather than the previous day’s meals: the body takes weeks to months to respond to changes in caloric intake and nutrient composition (Barnard, 1991; Tello, 2020; Vevers, 2024), at a pace that varies with metabolism and lifestyle.

Self-reported health (SRH) is the respondent’s rating of their own general health on the NHANES five-point scale, coded here so that higher values mean worse perceived health (“excellent” = 1 through “poor” = 5). SRH earns its place among the outcomes because it predicts metabolic conditions (Liu et al., 2017), morbidity, and mortality (Franks et al., 2003; Idler & Angel, 1990), and tracks healthy behaviors (Ware et al., 2022), all while measuring something the biomarkers cannot, the respondent’s own assessment (Bombak, 2013). Being self-reported, it also carries measurement error that reduces precision (Bound, 1991; Evans & Garthwaite, 2014). The objective measures described below mitigate that concern, and the SRH scores are standardized to remove scale effects, which is why the results refer to the outcome as adjusted SRH.

3.2.1 Measures of Physical Health

The objective measures come in two groups: anthropometric measures taken at the examination, and the blood-pressure readings and blood panel drawn in the laboratory.

Body Measures BMI is the workhorse measure of general population health, and pairing it with waist circumference sharpens what it says about obesity and related risks.² The two also combine into “A Body Shape Index” (ABSI),³ which is highly correlated with cardio-metabolic risk (Bertoli et al., 2017) and, once standardized by age and sex, identifies mortality hazards (Krakauer & Krakauer, 2012).

²See the CDC’s “Body Mass Index: Frequently Asked Questions” on the limitations of BMI as a stand-alone measure.

³ $\text{ABSI} = \text{WC} / (\text{BMI}^{2/3} \times \text{Height}^{1/2})$, where WC is waist circumference.

Table 2: Summary Statistics - Body Shape Variables

	Females			Males		
	22-44	45-64	65+	22-44	45-64	65+
BMI	28.6 (8)	29.8 (7.7)	28.8 (6.5)	28.5 (6.6)	29.3 (5.9)	28.7 (5.2)
Waist Circumference	93.6 (17.6)	98.5 (16.2)	98.6 (13.9)	98.3 (16.2)	104.5 (15.2)	106.1 (13.5)
ABSI	0.786 (1.002)	0.78 (0.998)	0.792 (0.994)	0.795 (0.996)	0.781 (0.999)	0.781 (0.992)

This table shows descriptive characteristics related to physical measures of individuals from the sample composed by pooling and filtering the NHANES waves from 2005-06 to 2017-18. Standard deviations are shown in parentheses.

ABSI stands for A Body Shape Index, calculated as $ABSI = \frac{WaistCirc}{BMI^{\frac{2}{3}} \times Height^{\frac{1}{2}}}$.

Table 2 summarizes the body-shape variables. Waist circumference rises monotonically with age for both genders, while BMI peaks in midlife and returns near its young-adult level by 65 and older, a divergence the index discussion below returns to.

Blood Pressure and Blood Panel Following Evans and Garthwaite (2014) and Seeman et al. (2008), seven clinical measures enter the indexes: systolic and diastolic blood pressure (mm Hg), pulse rate (bpm), total cholesterol (mg/dL), HDL concentration (mg/dL), glycated hemoglobin or HbA1c (%), and triglycerides (mg/dL).

Blood pressure and pulse rate capture cardiovascular health. Both blood-pressure readings track diets high in sodium and processed foods and low in potassium (Savica et al., 2010), and uncontrolled hypertension can lead to heart disease, stroke, and kidney disease.⁴ NHANES records up to four sequential blood-pressure readings per participant. I use the average of the second through fourth readings, discarding the first to mitigate “white-coat” bias (following the NHANES Manual of Operations) and excluding readings flagged as invalid (enhancement code 8). Pulse responds to diet indirectly through blood pressure, and elevated resting rates (above 90 bpm, American Heart Association, 2025) predict coronary heart disease (Gillum et al., 1991).

Total cholesterol, HDL cholesterol, HbA1c, and triglycerides capture metabolic health. Total cholesterol sums all forms in the blood, including LDL and HDL, while HDL (the “good” cholesterol) helps clear the other forms from the bloodstream. High total cholesterol paired with low HDL raises cardiovascular risk; saturated and trans fats push LDL up, while fiber and unsaturated fats push HDL up (Mensink et al., 2003). HbA1c measures average blood glucose over the previous two to three months, and levels of 6.5% or above (National Center for Biotechnology Information, 2025) indicate poor glycemic control and diabetes risk, driven over time by refined carbohydrates and

⁴<https://www.heart.org/en/health-topics/high-blood-pressure/health-threats-from-high-blood-pressure>

sugars (Boeing et al., 2000). Triglycerides are a blood fat whose elevation raises metabolic-syndrome risk, again driven by sugars, refined carbohydrates, and unhealthy fats (Austin et al., 1998).

Appendix Tables A.2 and A.3 report measurement units, observation counts, sample means, ideal ranges, and risk-group shares for each biomarker, split by whether the individual consumes any FAFH calories. The ideal ranges follow the American Heart Association and American Diabetes Association guidelines: systolic blood pressure below 130 mm Hg, diastolic below 90 mm Hg (National Heart, Lung, and Blood Institute, 2025), pulse below 90 bpm, total cholesterol below 200 mg/dL, HDL above 60 mg/dL, triglycerides below 150 mg/dL (Johns Hopkins Medicine, 2025), and HbA1c below 6.5%.

3.2.2 Indexes

The biomarkers combine into three indexes, each the average of its components' within-gender z-scores, following Kling et al. (2007) and Hoynes et al. (2016), and sign-adjusted so that worse health corresponds to more positive values. The metabolic index covers total cholesterol, HDL cholesterol, glycated hemoglobin, and triglycerides; the cardiovascular index covers systolic and diastolic blood pressure and pulse; and the body-shape index covers BMI and waist circumference. Adjusted self-reported health is the fourth main outcome. Aggregating biomarkers this way (following Seeman et al. (2008) and Evans and Garthwaite (2014)) improves statistical power against effects that move several markers of a domain in the same direction (Kling et al., 2007).

Because averaging correlated z-scores shrinks dispersion, the indexes are unit-free but do not have unit variance: the reported standard deviations run from about 0.57 to 0.89 across domains and age groups (Table 3), and only adjusted SRH has a standard deviation near one. Each index is also the mean of whichever components are non-missing for a given respondent, so composition varies across the sample. This matters most for the metabolic index, since triglycerides are drawn from the fasting subsample and cover roughly 47% of that index's respondents; for the remainder the metabolic index is a three-component average.

Table 3 shows the cardiovascular and metabolic indexes rising from young adulthood into midlife for both genders, indicating decline in both domains with age. The body-shape index also rises through midlife but improves slightly in the oldest group, the same pattern visible in the raw BMI series and consistent with selective survival among healthier individuals. Adjusted SRH rises slightly with age for both genders as well, meaning modestly worse self-rated health. Midlife is thus the critical period for these declines, while older ages likely reflect both biological aging and survivorship.

3.2.3 Healthy Eating Index

The Healthy Eating Index (HEI) is a 100-point USDA summary index of diet quality that evaluates how closely intake aligns with the federal Dietary Guidelines for Americans (DGA), scored from adequacy components (total fruit, whole grains, dairy, seafood, plant proteins) and moderation components (added sugars, sodium, refined grains, saturated fats) on 0–5 or 0–10 sub-scales, with

Table 3: Summary Statistics - Outcome Variables

	Females			Males		
	22-44	45-64	65+	22-44	45-64	65+
Cardiovascular Index	-0.026 (0.58)	0.184 (0.642)	0.136 (0.692)	0.079 (0.605)	0.248 (0.656)	0.088 (0.677)
Metabolic Index	-0.132 (0.59)	0.165 (0.708)	0.149 (0.599)	0.03 (0.63)	0.235 (0.695)	0.089 (0.569)
Body Shape Index	0.319 (0.887)	0.508 (0.845)	0.438 (0.719)	0.438 (0.832)	0.627 (0.743)	0.613 (0.66)
Adjusted SRH	-0.168 (0.953)	-0.021 (1.016)	0.015 (1.012)	-0.183 (0.943)	-0.033 (1.021)	0.002 (1.02)

This table shows descriptive characteristics related to physical measures of individuals from the sample composed by pooling and filtering the NHANES waves from 2005-06 to 2017-18. Standard deviations are shown in parentheses.

higher totals indicating better alignment. The index has three revisions (HEI-2005, HEI-2010, HEI-2015), each calibrated to its contemporary DGA. Two of them are computable from the food-pattern equivalents files available for these cycles, so cycles D (2005–2006) through F (2009–2010) are scored with HEI-2010 and cycles G through J (2011–2018) with HEI-2015. The scoring follows the published National Cancer Institute component definitions and cut points (National Cancer Institute, Division of Cancer Control and Population Sciences, 2023), re-implemented in R.

The HEI for each respondent is built from the first 24-hour recall, consistent with the treatment construction in Section 3.1. Component intakes are converted into densities (per 1,000 kcal for adequacy components and per kcal for moderation components), mapped through the scoring algorithm, and summed into a total on the 0–100 scale. Unlike the biomarker indexes, the HEI measures dietary quality rather than physiological health, and is best read as a complementary outcome that captures the channel through which FAFH consumption is thought to shape long-run health.

Women score higher than men on most adequacy components (total and whole fruits, total vegetables, greens and beans, dairy) and are almost indistinguishable on the moderation components (refined grains, sodium, added sugars, saturated fats); total protein foods is the one adequacy component that runs the other way. Figure B.1 in Appendix B shows the component comparison, and the gender gap in overall diet quality concentrates in what people add to their plate rather than in what they restrict. This baseline difference is useful context for the gendered FAFH effects in Section 5.

Figure 1 compares mean HEI component scores for respondents with any FAFH calories on the recall day against those with zero. The contrast runs in the same direction for both genders, and

it runs almost entirely one way: respondents who report no FAFH score higher on nearly every axis. They do better on the adequacy side (total and whole fruits by 7 to 9 points on the 0–100 normalized scale, whole grains by 6, dairy and total vegetables by 1 to 3) and on the moderation components where FAFH is supposed to do its damage (added sugars by 5 to 7 points, saturated fats by about 5). Only total protein foods favors FAFH consumers, and the sodium difference is negligible. Non-FAFH respondents are therefore not following a narrower diet; they are eating a better one on almost every dimension the HEI scores.

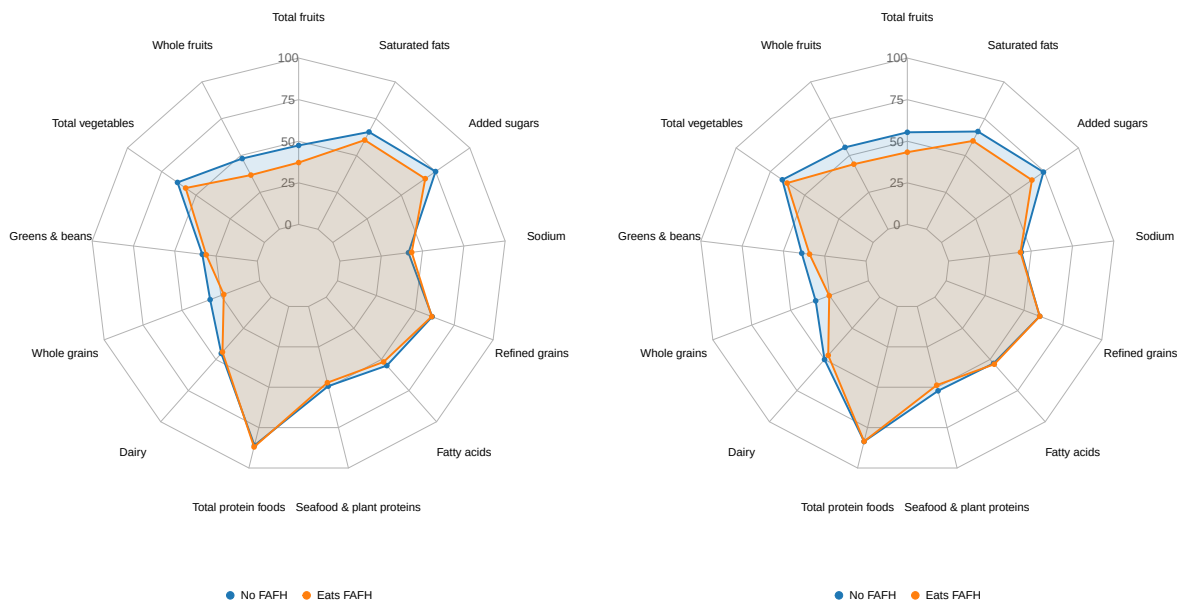


Figure 1: HEI-2015 component scores for the male sample (left panel) and female sample (right panel), separated by whether the respondent reports any FAFH calories on the recall day.

Data source: NHANES 2011–2018.

Notes: “Eats FAFH” = $C_i > 0$ on the recall day; “No FAFH” = $C_i = 0$. Each axis is the mean component score normalized to the 0–100 scale (percent of HEI-2015 component maximum)

That across-the-board gap is exactly what motivates the bunching design in Section 4. People who consume zero FAFH differ from FAFH consumers on every component of the HEI at once, which is the signature of selection on something broader than the food itself: the household food environment, dietary preferences, and time use that jointly determine both whether someone eats out and what they eat when they do not.

3.3 Demographic Variables

Table 4 presents demographic characteristics across age groups for females and males.

The White share rises with age for both genders, reflecting broader population trends over the sample period. Marriage rates climb from young adulthood into midlife for both, then diverge

Table 4: Summary Statistics - Demographic Variables

	Females			Males		
	22-44	45-64	65+	22-44	45-64	65+
White	0.606 (0.487)	0.7 (0.488)	0.8 (0.496)	0.604 (0.489)	0.719 (0.492)	0.806 (0.493)
Married	0.48 (0.498)	0.614 (0.499)	0.466 (0.489)	0.447 (0.498)	0.672 (0.477)	0.735 (0.468)
College Complete	0.309 (0.44)	0.305 (0.42)	0.2 (0.358)	0.276 (0.419)	0.296 (0.423)	0.322 (0.418)

This table shows demographic characteristics of individuals from the sample composed by pooling and filtering the NHANES waves from 2005-06 to 2017-18. Standard deviations are shown in parentheses.

sharply: they fall from 0.61 to 0.47 among females aged 65 and older, largely through widowhood, while continuing to rise to 0.74 among males. College completion moves in opposite directions across the age range as well, dropping by a third among the oldest females (from 0.31 to 0.20) while edging up across male age groups, a cohort pattern in who had access to higher education.

These variables serve as the controls throughout the analysis: age and its square, the family poverty-income ratio, a White indicator, dummies for being married and holding a college degree, and full indicator sets for education, marital status, special-diet status, and survey wave, along with indicators flagging missing values so that incomplete records stay in the sample. Section 4.2 describes how the controls enter the estimation, both directly and through the hierarchical clustering that operationalizes the conditional form of the identifying assumptions.

3.4 Stylized Facts on Dietary Patterns

Four patterns set the stage for the analysis: where American calories come from, how food spending is shifting, how FAFH evolves over the life cycle, and how it is spread across meals.

Figure 2 breaks average daily caloric intake out by food source: home, restaurants, fast food, cafeterias, and food assistance programs (FAP). Item by item, fast food packs the most calories (Figure 2a), while items from the other sources are roughly comparable. Summed over the day, though, home cooking dominates (Figure 2b): FAFH supplies the largest share of total intake despite its lighter items, simply because people eat it more often. Both panels are roughly flat across cycles, so dietary patterns by source have changed little over the period.

Figure 3 turns from calories to dollars, plotting monthly family food expenditure in 2020 dollars from the NHANES 30-day spending questions,⁵ weighted to be nationally representative. Spending rose in absolute terms across every category, groceries and eating out alike, with the largest move-

⁵https://www.cdc.gov/Nchs/Nhanes/2007-2008/CBQ_E.htm

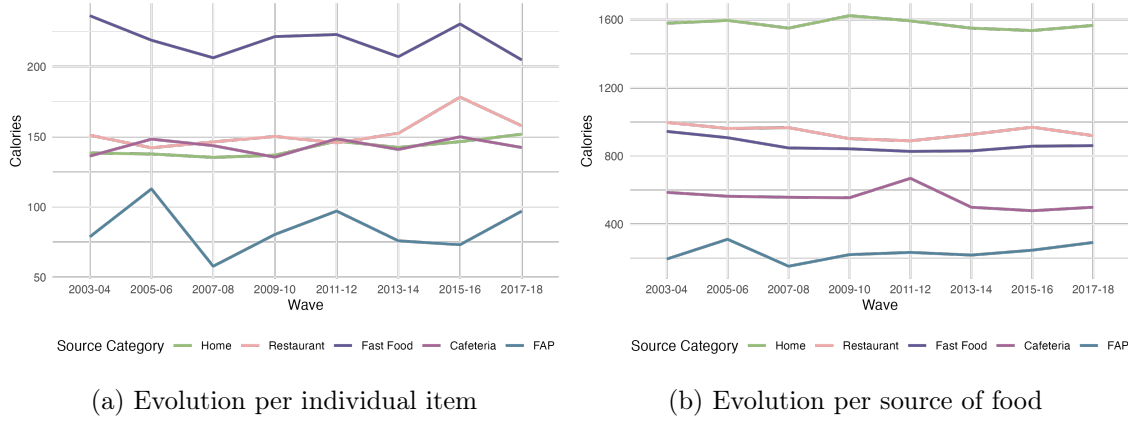


Figure 2: Evolution of average daily calories consumed in the United States across NHANES cycles, broken out by the source of food.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: Figure 2a represents the average caloric amount of an individual item consumed. Figure 2b represents the average caloric amount consumed from each source.

The figures exclude items categorized as “Other” (see Table A.1 in the Appendix for details).

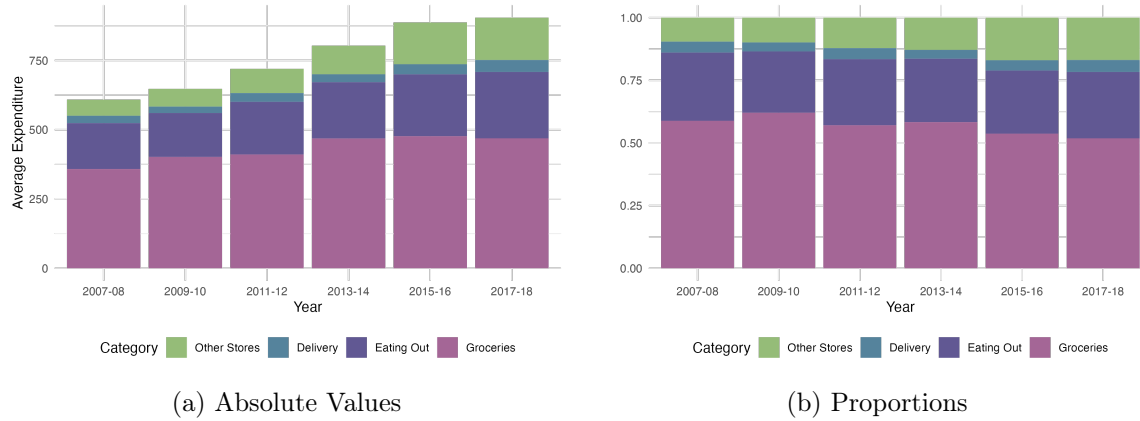


Figure 3: Evolution of average monthly family food expenditure by category, reported both in inflation-adjusted absolute values and as shares of total food spending.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Values adjusted for 2020 US\$, and not including alcoholic beverages.

The graph shows the evolution of family expenditure from 2007-08 to 2017-18 in both absolute values and proportions. The categories account for all money spent on food in 30 days.

ment in the “other sources” category (which the codebook leaves undocumented). The shares in the right panel move much less than the levels, so the growth reflects rising food prices and budgets more than a reallocation between eating at home and eating out.

Over the life cycle, the two sources behave very differently (Figure 4). FAH intake holds roughly steady from age 22 to 80, while FAFH peaks in the twenties and declines steadily thereafter. Younger adults eat more overall, and the life-cycle decline in total intake comes almost entirely from food eaten away from home.

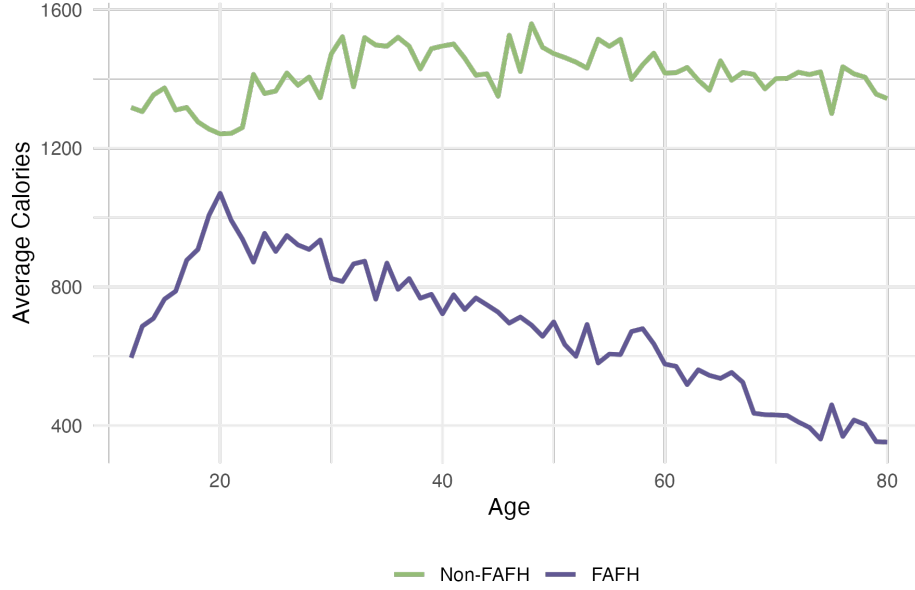


Figure 4: Average daily calorie consumption over the life cycle, separated by FAFH and Non-FAFH sources.

Data source: National Health and Nutrition Examination Survey (NHANES).

Tables 5 (females) and 6 (males) compare respondents who eat only FAFH with those who consume any Non-FAFH. The first column covers exclusive-FAFH consumers, the second those with positive Non-FAFH calories, the third reports the p-value of a t-test between the two, and the fourth summarizes the table’s full sample (the descriptive tables use different filters from the estimation; see the reconciliation note in Section 3). Non-FAFH consumers take in more total calories than exclusive-FAFH consumers at every age, with the gap larger for males, and the differences are statistically significant throughout.

Finally, Figure 5 splits consumption by meal type. Breakfast is the lightest meal and dinner the heaviest. Snack calories run higher at home than away, consistent with the easier access to snacks in one’s own kitchen, and drink calories are non-trivial in both.

4 Empirical Strategy

The empirical strategy identifies the causal effect of Non-FAFH calories on health from the bunching of respondents at zero consumption, following Caetano et al. (2024a) and its application to maternal labor supply in Caetano et al. (2024b). The treatment C is the individual’s daily Non-FAFH caloric intake (Section 3.1), and a large share of respondents reports exactly zero. Standard identification tools have little traction in this setting: no institution assigns restaurant meals, no credible instrument shifts Non-FAFH intake without touching health, and the cross-sectional structure of NHANES rules out individual fixed effects. The bunching design instead runs on a feature the data already have, the mass of non-consumers at zero.

Table 5: Summary Statistics - Females

	FAFH = 0	FAFH >0	<i>p-value</i>	Total
22-44				
Number of Individuals	1,001	5,427	-	6,428
Calories Consumed (FAH)	1,648	1,146	0.000	1,222
	(644)	(608)		(643)
Calories Consumed (FAFH)	-	750	-	636
		(569)		(592)
Total Calories Consumed	1,648	1,896	0.000	1,858
	(644)	(677)		(677)
45-64				
Number of Individuals	1,069	4,788	-	5,857
Calories Consumed (FAH)	1,604	1,181	0.000	1,252
	(578)	(567)		(586)
Calories Consumed (FAFH)	-	614	-	511
		(476)		(491)
Total Calories Consumed	1,604	1,795	0.000	1,763
	(578)	(601)		(605)
65+				
Number of Individuals	1,214	3,170	-	4,384
Calories Consumed (FAH)	1,512	1,143	0.000	1,239
	(505)	(503)		(528)
Calories Consumed (FAFH)	-	496	-	366
		(400)		(401)
Total Calories Consumed	1,512	1,638	0.000	1,605
	(505)	(534)		(529)

This table shows descriptive characteristics of females from the sample composed by pooling and filtering the NHANES waves from 2005-06 to 2017-18. Standard deviations are shown in parentheses.

The maintained assumption is that the outcome is continuous in the treatment at zero: consuming a token amount of FAFH, a small candy on the way home, cannot change health discontinuously. The assumption is untestable, since negative FAFH calories are unobservable, but it is hard to argue against at this margin. Its power is what it implies for the data: under continuity, any jump in average outcomes at exactly zero FAFH cannot be an effect of the food, so it must reflect who sits at zero. One caveat is worth stating up front: the treatment is a single day's intake, so interpreting zero FAFH on the recall day as a stable low-FAFH lifestyle is an extrapolation, and the measured

Table 6: Summary Statistics - Males

	FAFH = 0	FAFH > 0	<i>p-value</i>	Total
22-44				
Number of Individuals	750	5,673	-	6,423
Calories Consumed (FAH)	2,396	1,523	0.000	1,619
	(1,038)	(883)		(943)
Calories Consumed (FAFH)	-	1,153	-	1,026
		(850)		(880)
Total Calories Consumed	2,396	2,676	0.000	2,646
	(1,038)	(976)		(988)
45-64				
Number of Individuals	1,023	4,720	-	5,743
Calories Consumed (FAH)	2,211	1,570	0.000	1,667
	(881)	(807)		(850)
Calories Consumed (FAFH)	-	905	-	769
		(689)		(708)
Total Calories Consumed	2,211	2,475	0.000	2,436
	(881)	(847)		(860)
65+				
Number of Individuals	1,255	3,264	-	4,519
Calories Consumed (FAH)	1,870	1,445	0.000	1,546
	(639)	(671)		(683)
Calories Consumed (FAFH)	-	678	-	517
		(521)		(524)
Total Calories Consumed	1,870	2,122	0.000	2,062
	(639)	(697)		(691)

This table shows descriptive characteristics of males from the sample composed by pooling and filtering the NHANES waves from 2005-06 to 2017-18. Standard deviations are shown in parentheses.

intake is best read as a proxy for the propensity to eat FAFH.

Section 4.1 presents the bunching evidence and shows that the respondents at zero differ systematically, in observables and unobservables alike, from those consuming marginally positive amounts. Section 4.2 converts that evidence into an estimator through a control-function representation, states the identifying assumptions, and describes inference.

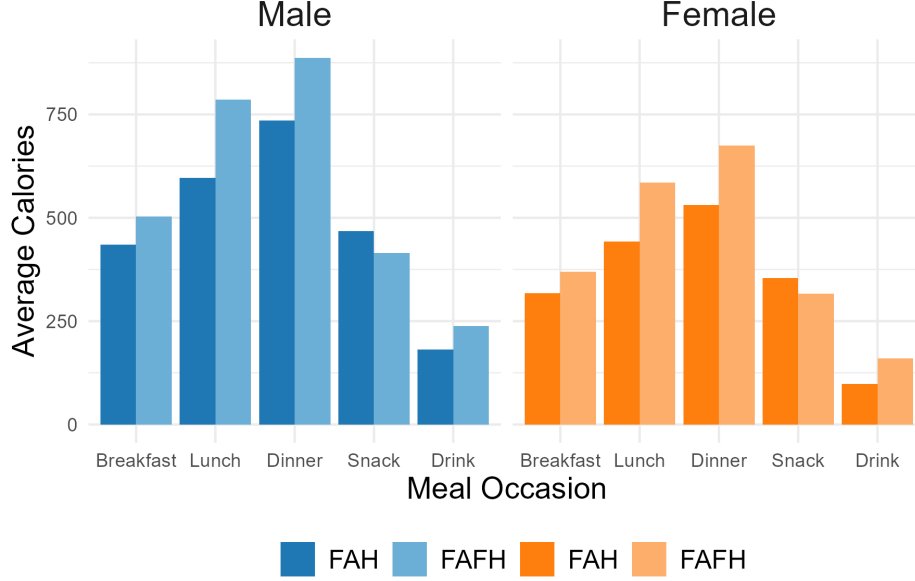


Figure 5: Average calorie consumption by meal type, broken out by FAH and FAFH sources.

Data source: National Health and Nutrition Examination Survey (NHANES).

4.1 Selection on Unobservables

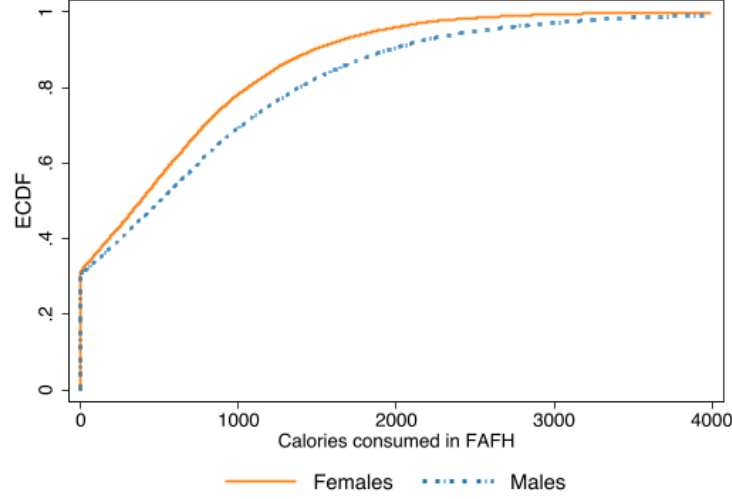
The identification strategy starts from the bunching itself. Figure 6a plots the empirical CDF of C , and approximately 30% of individuals report zero FAFH calories. The share ranges between 20% and 40% across age groups within gender (Figures 6b and 6c), with younger individuals consuming more FAFH than older ones, consistent with Saksena et al. (2018).

The mass at zero matters because the individuals in it are not one type but two. Those with positive FAFH calories are at an interior solution: they consume exactly the amount they want. At zero, some respondents are just indifferent at the margin and would consume a little FAFH if nudged, while others would consume less than zero if they could, declining a restaurant meal even when it is fully paid for, and are held at zero only by the physical constraint that calorie consumption cannot be negative. The stronger that underlying opposition, the further the individual sits from marginal indifference, yet everyone in this group is observed at the same value.

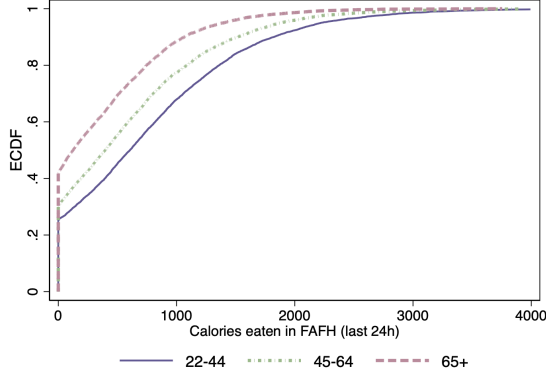
To formalize this, introduce a selection variable C^* representing the individual's type, capturing all confounders (observed and unobserved) that influence C . For $C > 0$, $C^* = C$; for $C = 0$, $C^* \leq 0$. While C cannot be negative, C^* has no such restriction,

$$C = C^* \cdot \mathbf{1}(C^* \geq 0), \text{ with } \mathbb{P}(C^* < 0) > 0, \quad (1)$$

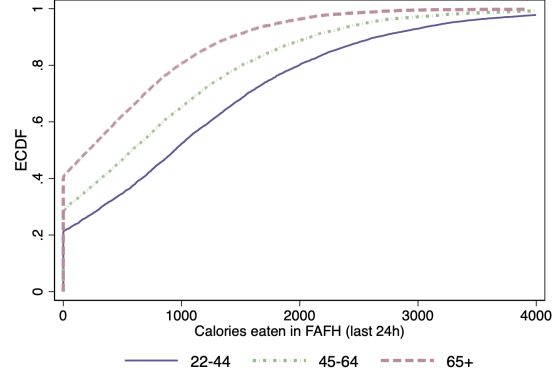
where the condition $\mathbb{P}(C^* < 0) > 0$ guarantees that some individuals are at the corner, which is



(a) Total sample



(b) Females, by age groups



(c) Males, by age groups

Figure 6: Empirical CDF of daily calories from FAFH, showing bunching at zero and the share of respondents above the 2,500-kcal threshold.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: The vertical dashed line marks 2,500 daily FAFH calories and the horizontal dashed line marks the 90% cumulative density: in every gender and age group, more than 90% of the sample consumes less than 2,500 FAFH calories.

what produces the bunching. Since $C = \max\{0, C^*\}$, average type is discontinuous at zero,

$$\Delta_{C^*} = \lim_{c \downarrow 0} \mathbb{E}[C^* \mid C = c] - \mathbb{E}[C^* \mid C = 0] = 0 - \mathbb{E}[C^* \mid C = 0] = -\mathbb{E}[C^* \mid C = 0] > 0. \quad (2)$$

As consumption approaches zero from above, average type approaches zero with it, but the bunched group's average type $\mathbb{E}[C^* \mid C = 0]$ is strictly negative. The gap measures how far, on average, the bunched group sits from marginal indifference.

The discontinuity in C^* is unobservable, but if it is real, variables correlated with type should jump at $C = 0$, and they do (Figure 7).

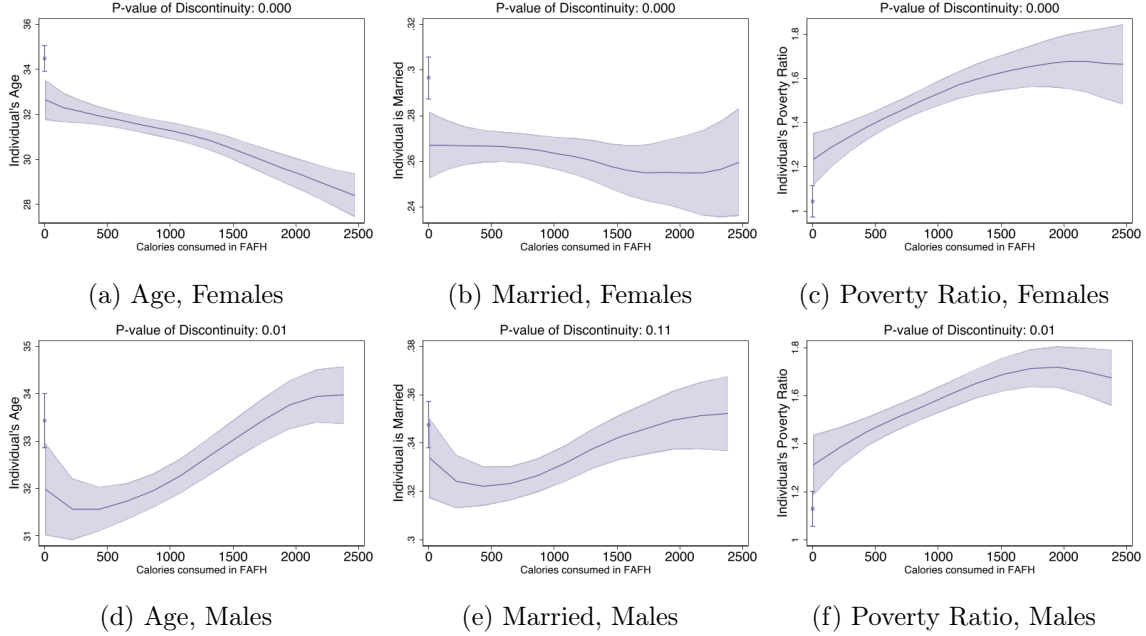


Figure 7: Discontinuities in observable demographic characteristics (age, marital status, poverty ratio) at the FAFH bunching point, separately by gender, as a placebo check for the identification.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Confidence intervals of 95% are shown on the plot, and the p-value of the discontinuity at $C = 0$ is shown on top of each panel. The bandwidth used for the construction of each plot is of 500 calories.

The income-poverty ratio is a number greater than zero that represents the ratio between the family annual income and the poverty level. A ratio less than 1 means that the income is less than the poverty level.

The discontinuity plots are displayed over the range from 0 to 2,500 FAFH calories, which covers more than 90% of the sample in every gender and age group; the sparse tail beyond 2,500 is not shown.

The bunched respondents differ from the lightest consumers in a consistent direction. They are about one year older on average (both genders, $p = 0.03$), and they are substantially poorer, with a poverty-income ratio of 1.02 against 1.16 for marginal female consumers ($p = 0.01$) and 1.09 against 1.34 for males ($p < 0.01$). Bunched females are also slightly more likely to be married (0.28 against 0.27, $p = 0.02$), while the marital gap for males is not significant ($p = 0.28$). Away from zero, the profiles behave as the dietary patterns of Section 3.4 would predict: income rises with FAFH intake over most of the range for both genders, and age falls with intake for females, while the male age profile is U-shaped. Under treatment-effect continuity, a marginal calorie of FAFH cannot cause someone to be older, poorer, or married, so these discontinuities can only mean that the people at zero are a selected group.

Age, income, and marital status are observable and can be controlled for. The concern is selection on unobservables, so Figures 8 (females) and B.2 (males, appendix) repeat the analysis on the outcome variables themselves.

For females, the jump at zero runs the same way in all four panels: the bunched group's average

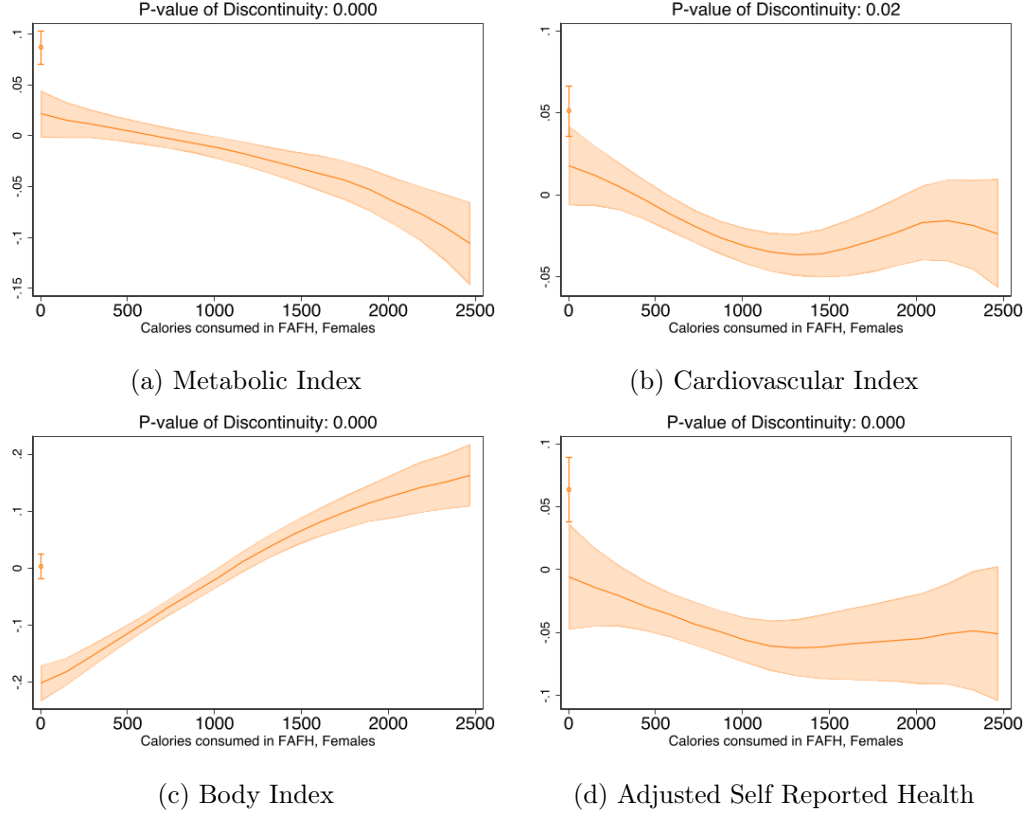


Figure 8: Evidence of selection at the bunching point for the female sample, using raw (unresidualized) outcomes against FAFH calories.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Confidence intervals of 95% are shown on the plot, and the p-value of the discontinuity at $C = 0$ is shown on top of each panel.

The plot represents the effect of an additional 1,000 calories on each outcome variable, regressed on zero, and a bandwidth of 500 calories for other values.

Adjusted self-reported health is calculated by standardizing the self-reported health by wave.

outcome sits above the marginal consumers', meaning worse health at the mass point. The gap is largest for the body-shape index (0.00 against -0.20 , $p < 0.01$), clear for the metabolic index (0.09 against 0.02, $p < 0.01$) and adjusted SRH (0.06 against -0.01 , $p < 0.01$), and smallest, though still significant, for the cardiovascular index (0.05 against 0.02, $p = 0.02$). The slopes away from zero, in contrast, differ by outcome: the metabolic index falls as intake rises, so in the raw cross-section the heaviest female FAFH consumers are the metabolically healthiest, while the body-shape index rises steeply. Taken at face value, the metabolic panel would say that eating out improves metabolic health. Taken seriously, the uniform direction of the jumps says that health-relevant characteristics select people into and out of FAFH, and that the slopes mix this selection with any true effect in proportions that differ outcome by outcome.

Since observable characteristics could explain part of these jumps, Figure 9 repeats the exercise on residualized outcomes, holding the control variables and clusters fixed so they cannot account

for any part of the discontinuity.

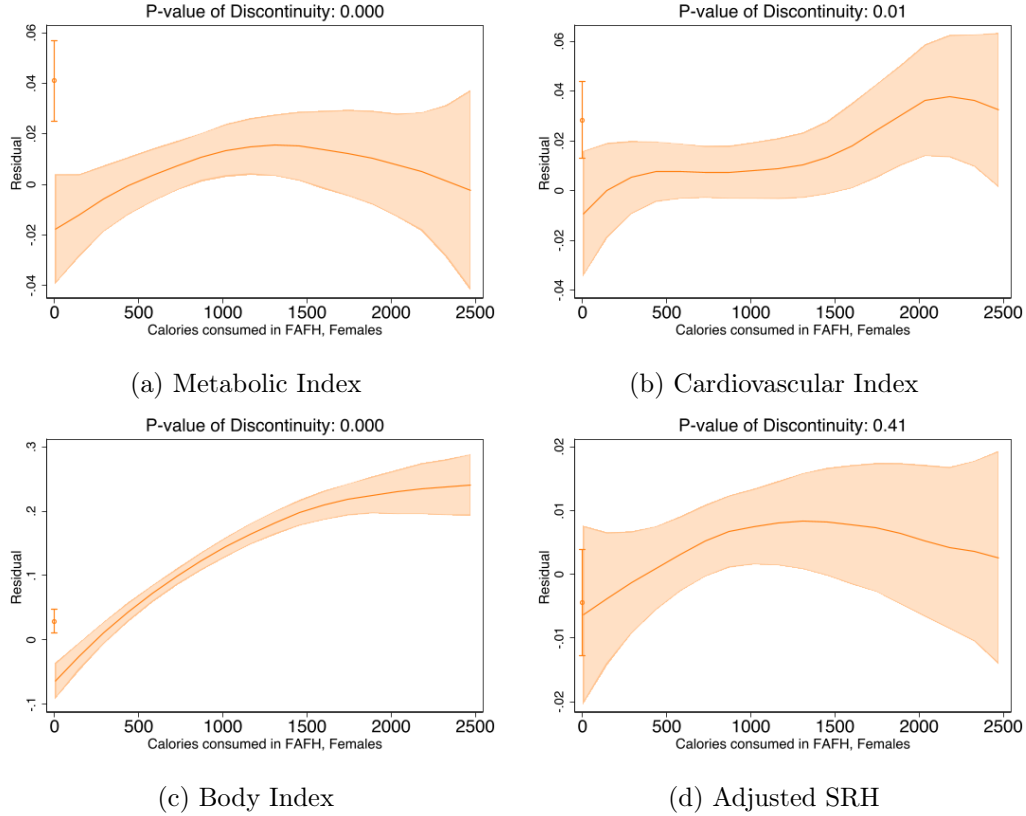


Figure 9: Evidence of selection at the bunching point for the female sample, using residualized outcomes (net of demographic controls) against FAFH calories.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The covariates on this model includes: age, age squared, poverty ratio, and dummies for education, wave, marital status, gender, and whether the individual is dieting or not, using a bandwidth of 500 calories. Confidence intervals of 95% are shown on the plot, and the p-value of the discontinuity at $C = 0$ is shown on top of each panel. Adjusted self-reported health is calculated by standardizing the self-reported health by wave.

The discontinuities survive residualization. The bunched group's residual sits above the marginal consumers' for the metabolic index (0.04 against -0.02 , $p < 0.01$), the cardiovascular index (0.03 against -0.01 , $p = 0.01$), and the body-shape index (0.03 against -0.07 , $p < 0.01$), evidence of selection on unobservables in the same direction as the observable selection. The adjusted-SRH gap, by contrast, disappears entirely once controls are added ($p = 0.41$): whatever separates non-consumers from consumers in how they rate their health, the observables capture it. The slopes are just as informative, and residualization treats them very differently: the metabolic and SRH gradients nearly vanish, so most of their raw slopes reflected composition rather than consequence; the cardiovascular gradient flattens to a mild positive drift; and the body-shape gradient remains steep, anticipating the result in Section 5 that most of the body-shape association survives the correction.

The slope of the residualized outcome at any $C = c > 0$ mixes the treatment effect with whatever selection remains, while the jump at zero is selection alone.⁶ If selection enters with a constant slope over the whole range of latent demand, the condition Section 4.2 imposes, then the selection measured at the bunching point extends to the rest of the distribution, and the treatment effect is the observed slope purged of it. For adjusted SRH in females this logic already implies a near-zero treatment effect (Figure 9d), a preview of the formal estimates.

The corresponding analysis for males is in the appendix (Figures B.2 and B.3), and the pattern is weaker throughout. The raw male jumps are visible for the cardiovascular index (0.06 against 0.03, $p = 0.07$) and the metabolic index (0.02 against -0.01 , $p = 0.03$), but once controls are added neither gap survives cleanly (cardiovascular $p = 0.10$, metabolic $p = 0.17$): for males, the observables account for most of what separates non-consumers from consumers. Selection on unobservables in the physiological outcomes is therefore largely a female phenomenon in this setting, which matches the near-zero male selection terms the estimation recovers for those outcomes in Section 5; diet-quality selection (the HEI) is the exception, present for both genders.

4.2 Identification Strategy

Let H denote an individual’s health measure. It depends linearly on the treatment and on unobserved factors U ,

$$H = \alpha + \beta C + U, \tag{3}$$

where β is the parameter of interest and U collects everything else that determines health. The endogeneity problem is that U is correlated with C : since C^* is by construction a sufficient index of all confounders, the concern is encapsulated in how U varies with C^* . Identification begins by restricting this dependence.

Assumption 1 (Conditional linearity in C^*). $\mathbb{E}[U \mid C^*, X] = m(X) + \delta \cdot (C^* - \mathbb{E}[C^* \mid X])$.

In words, the conditional mean of the unobservable health determinants moves linearly with latent FAFH demand, once the observed controls are accounted for. The level term $m(X)$ lets mean health differ freely across covariate cells; the substantive restriction is on the slope: the unobserved factors that lead a person to eat out more, whether the household food environment, time constraints, or dietary norms, must enter with a constant slope δ throughout the support of C^* , not merely operate in the same direction. When would the conditional mean bend rather than run straight? One threat is that zero-FAFH respondents mix opposing types: committed home cooks whose unobservables predict good health, and homebound or ill individuals whose unobservables predict poor health. This threatens the assumption only to the extent that it makes $\mathbb{E}[U \mid C^*, X]$ non-linear in C^* , since the assumption restricts the average of U at each level of latent demand, not the behavior of any single household. The concern is further limited because the assumption needs to hold only conditional on X : the controls and clusters absorb the observable part of health-consciousness, income, and age through $m(X)$, so linearity is required only within

⁶Section 4.2 formalizes this decomposition.

groups of observationally similar individuals. The derivation below specializes to the case where $m(X)$ and $\mathbb{E}[C^* | X]$ are constant, so $\mathbb{E}[U | C^*] = m + \delta(C^* - \mu)$ with the constants absorbed into the intercept; the Controls discussion at the end of this section restores the conditional form. The approximately linear residualized profiles in Figures 9 and B.3 support the linearity, and Sections 6.1 and 6.3 probe it by varying the distributional assumption and the clustering configuration.⁷

Under Assumption 1, Equation 3 becomes

$$H = \alpha + \beta C + \delta C^* + \varepsilon, \quad \mathbb{E}[\varepsilon | C, C^*] = 0, \quad (4)$$

so β and δ would be identified by an ordinary regression of H on a constant, C , and C^* , if C^* were observable. It is not, and the bunching design constructs a control function that substitutes for it. Using the identity $C^* = C + C^* \cdot \mathbf{1}(C = 0)$, which holds because $C^* = C$ on the positive side, and taking conditional expectations of Equation 4 given C ,

$$\mathbb{E}[H | C] = \alpha + (\beta + \delta) C + \delta \mathbb{E}[C^* | C = 0] \cdot \mathbf{1}(C = 0). \quad (5)$$

Equation 5 is the empirical counterpart of the residualized-outcome plots in Section 4.1. The slope on the positive side identifies $\beta + \delta$, the sum of the causal effect and the selection effect. The discontinuity at zero contains no β at all, which is treatment-effect continuity doing its work. Define the outcome discontinuity $\Delta_H \equiv \lim_{c \downarrow 0} \mathbb{E}[H | C = c] - \mathbb{E}[H | C = 0]$ and recall the latent gap $\Delta_{C^*} = -\mathbb{E}[C^* | C = 0] > 0$ from Equation 2. Then Equation 5 implies

$$\Delta_H = \delta \cdot \Delta_{C^*}, \quad (6)$$

the selection slope scaled by the latent-demand gap, exactly the “jump at zero is selection alone” logic of Section 4.1. Both Δ_H and the positive-side slope are observed, so knowing the magnitude of $\mathbb{E}[C^* | C = 0]$ delivers δ from the ratio of the outcome discontinuity to the latent discontinuity, and β by subtraction from the slope. Equivalently, an OLS regression of H on a constant, C , and the constructed control function $C + \mathbb{E}[C^* | C = 0] \cdot \mathbf{1}(C = 0)$ recovers both parameters at once. The one remaining unknown is $\mathbb{E}[C^* | C = 0]$: how far below zero the bunched group’s latent demand sits, on average. It must be recovered from a restriction on the distribution of C^* , whose lower tail is never observed.

I implement three versions of that restriction. Two are semiparametric, positing a parametric shape for the latent distribution below zero, uniform or normal, and reading $\mathbb{E}[C^* | C = 0]$ off the fitted shape. The third, preferred version imposes no parametric family at all: it recovers the lower tail of C^* by reflecting the observed upper tail of the consumption distribution.

Assumption 2 (Tail Symmetry of C^*). *Let $\bar{s}$ denote the $(1 - \mathbb{P}(C = 0))$ -quantile of the distribution of C , including the mass point at zero. For $s > \bar{s}$, the density of C^* at $-s + \bar{s}$ equals the density of C at s : the distribution of latent demand below zero mirrors the observed distribution of consumption*

⁷The implementation also allows the selection slope to differ across clusters of X ($K_d > 1$ in the notation of Section 6.3); the baseline sets $K_d = 1$, a single δ , which is the specification reported in the results tables.

above $\bar{s}$.

By construction, the mass of consumption above $\bar{s}$ equals the bunching share $\mathbb{P}(C = 0)$, so the reflected tail carries exactly the mass of the bunched group, and the assumption specifies the entire distribution of C^* below zero from a part of the distribution that is observed. The shape of the observed middle range of consumption, between zero and $\bar{s}$, matters only through the location of $\bar{s}$. A closed-form expression for the unknown expectation follows,

$$\mathbb{E}[C^* \mid C = 0] = \bar{s} - \mathbb{E}[C \mid C \geq \bar{s}], \quad (7)$$

which is fully nonparametric in the observed positive-side distribution.⁸ Because the mirrored tail is the empirical tail rather than a fitted density, it is longer and flatter than a parametric alternative would be, so tail symmetry accounts for extreme latent-demand values to which a fitted normal assigns little mass. This makes it the conservative choice: it implies the most negative $\mathbb{E}[C^* \mid C = 0]$ of the three specifications, hence the smallest selection correction. The estimates bear this out: the selection term $\hat{\delta}$ shrinks in magnitude from the uniform to the tail-symmetric specification in every row with substantial selection (the female metabolic and cardiovascular indexes and the HEI for both genders), and the corrected $\hat{\beta}$ moves with it, while rows whose selection terms are near zero wander within their confidence intervals rather than following the ordering. Section 6.1 reports the full comparison.

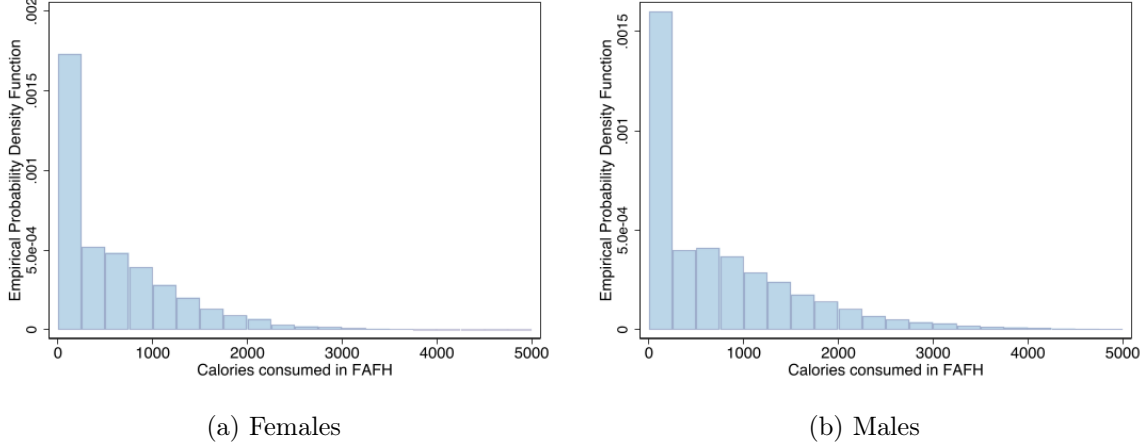


Figure 10: Empirical probability density of daily calories from FAFH, separately by gender, showing the mass point at $C = 0$ and the upper tail that the tail-symmetry construction reflects below zero.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: Waves 2005–06 through 2017–18, bins of 250 calories.

The signs in the results tables line up with this construction throughout. For females, the residualized outcomes at the bunching point sit above the positive-side limit for all three physiological

⁸The construction requires the bunching share to be strictly below one half, so that $\bar{s} > 0$, and a positive, continuous density of C in a neighborhood of $\bar{s}$, so that the quantile is unique. Both hold comfortably in this application, and in the implementation any cluster of X bunched above one half is switched to the normal (tobit) specification.

indexes (Section 4.1), so $\Delta_H < 0$; combined with $\Delta_{C^*} > 0$, Equation 6 implies $\hat{\delta} < 0$, which is what the female table reports in all three rows. Individuals with high latent FAFH demand carry unobservables associated with better health, so the naive slopes understate the harm, and the correction subtracts a negative selection slope, moving the estimates up from the naive ones. For adjusted SRH the residualized jump is indistinguishable from zero and so is $\hat{\delta}$, and for males, whose residualized jumps do not survive the controls, the estimated selection terms on the health indexes are all near zero. The HEI is the exception on the male side: its selection term is significant for both genders, so diet-quality selection into FAFH is present even where physiological selection is not. One magnitude caveat applies to the body-shape index: its residualized profile is concave, steep near the corner and flatter beyond roughly 1,500 calories, so the local jump visible in the figure implies a larger selection term than the global linear extrapolation the estimator uses ($\hat{\delta} = -0.05$). Estimates from narrow windows around the corner are accordingly larger, stabilizing as the window widens, the pattern Section 6.2 traces.

Controls

Adding control variables makes both assumptions conditional: linearity and tail symmetry are required to hold only within groups of observationally similar individuals, and the level differences across groups are absorbed through $m(X)$. Following Caetano et al. (2024a), the groups are formed by hierarchical clustering of the control variables, and the bunching share, the threshold $\bar{s}$, and the censored expectation of Equation 7 are computed within each cluster, so X enters the estimation both directly and through the clusters. Section 6.3 confirms consistency across clustering configurations.

Inference

Standard errors and confidence intervals come from a non-parametric bootstrap. Each iteration draws a sample of respondents with replacement (individual-level resampling, matching NHANES’s cross-sectional structure) and re-runs the full estimation pipeline: the hierarchical clustering of X , the control-function fit, and all five estimator variants in the results tables, namely the uncorrected regressions with and without controls, the semiparametric uniform, the semiparametric normal, and the nonparametric tail-symmetric specifications. Each coefficient’s standard error is the standard deviation of its bootstrap distribution, and the 95% confidence intervals in the figures are the corresponding ± 1.96 SE bands.

The headline estimation in Section 5 uses 250 bootstrap replications, the distribution-sensitivity robustness (Section 6.1) uses 100 replications, and the cluster-number robustness (Section 6.3) uses 25 replications at each K (or K_d), given the cost of sweeping K across 30 values for each of the five outcomes and two genders. Since each NHANES respondent appears once, individual-level bootstrap coincides with the standard non-clustered bootstrap, and no wave-level clustering is added since the wave fixed effects are absorbed via X .

5 Results

5.1 Main Results

Results are reported separately by gender to account for distinct metabolic responses. Table 7 covers females (main text) and Table A.4 covers males (appendix), with analogous interpretation.

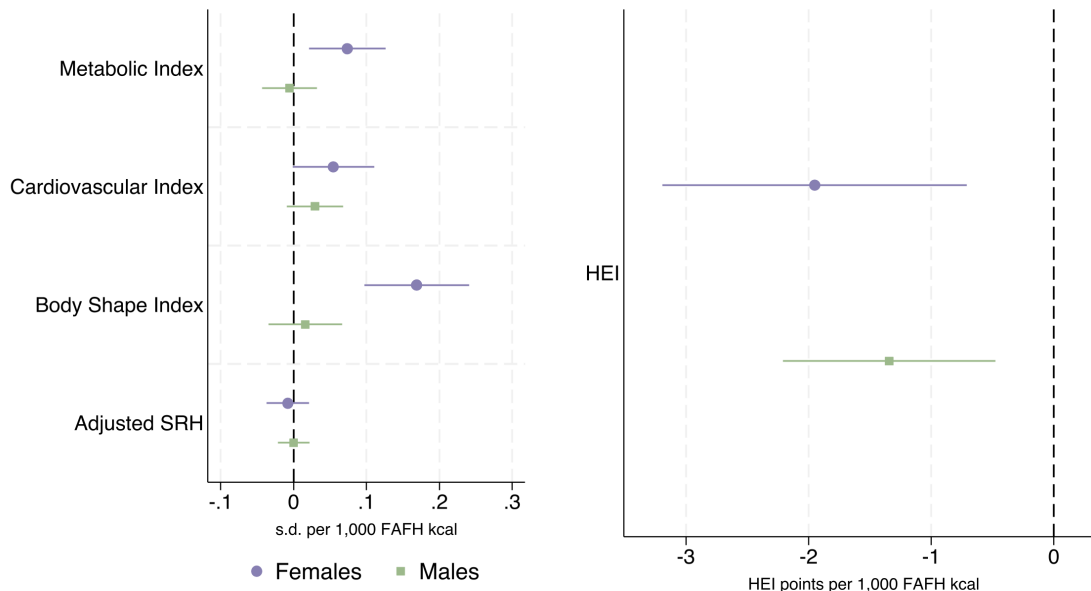


Figure 11: Tail-symmetric estimates of the effect of an additional 1,000 FAFH kcal/day, separately by gender: health indexes and adjusted SRH in standard deviations (left panel) and the HEI in points on its 0–100 scale (right panel).

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the results from regressions of the metabolic, cardiovascular, and body-shape indexes, adjusted SRH (left panel), and the HEI (right panel) on the amount of calories consumed in FAFH on the first dietary recall, using as controls the demographic characteristics and clusters, separating the sample between males and females. The HEI is plotted on its own axis because it is measured in points rather than standard deviations. Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

Figure 11 plots the tail-symmetric estimates. In the left panel, the coefficient on each index is the effect of an additional 1,000 FAFH calories, in standard deviations of the standardized index, where higher index values indicate worse health outcomes; the right panel reports the HEI effect in points, where lower values mean worse diet quality. For females, the physiological indexes rise with FAFH consumption, consistent with the literature on health deterioration as FAFH intake grows. For males, no health index responds significantly to an additional 1,000 FAFH calories; as shown below, the male response operates through diet quality instead.

Sample sizes vary by outcome. For females N ranges from 13,687 (metabolic index) to 14,314 (HEI and adjusted SRH); for males, from 14,639 to 15,274. All five outcomes come from the same analytic sample (NHANES cycles D through J, non-pregnant, typical eating day on the first recall), with per-outcome variation driven by missingness on

Table 7: Effects of Calories consumed in FAFH for Females

Dependent Variable		(1) Uncorrected No Controls	(2) Uncorrected w/ Controls	(3) Semip. Uniform	(4) Semip. Normal	(5) Nonp. Tail Symmetric
Metabolic Index (N= 13,687)	β	-0.067** (0.007)	-0.009 (0.007)	0.099** (0.039)	0.079** (0.028)	0.074** (0.027)
	δ			-0.096** (0.035)	-0.073** (0.024)	-0.068** (0.022)
Cardiovascular Index (N= 14,031)	β	-0.039** (0.007)	0.004 (0.007)	0.093** (0.041)	0.059** (0.030)	0.054* (0.029)
	δ			-0.079** (0.036)	-0.046* (0.025)	-0.042* (0.023)
Body Shape Index (N= 14,168)	β	0.086** (0.011)	0.108** (0.010)	0.160** (0.051)	0.167** (0.038)	0.169** (0.037)
	δ			-0.046 (0.044)	-0.049 (0.031)	-0.050* (0.029)
Adjusted SRH (N= 14,314)	β	-0.050** (0.012)	0.008** (0.004)	0.001 (0.021)	-0.003 (0.016)	-0.008 (0.015)
	δ			0.006 (0.019)	0.009 (0.013)	0.013 (0.012)
HEI (N= 14,314)	β	-4.641** (0.129)	-4.078** (0.150)	-1.388 (0.903)	-1.917** (0.661)	-1.950** (0.634)
	δ			-2.371** (0.783)	-1.788** (0.531)	-1.740** (0.502)

the specific biomarkers or HEI components.

Columns 1 and 2 report linear regressions that do not correct for endogeneity in FAFH consumption. Columns 3, 4, and 5 report control-function estimates under three distributional assumptions (semiparametric uniform, semiparametric normal, nonparametric tail-symmetric). Column 5 is the preferred specification (the least parametric one) and is shown in bold, and the discussion below refers to it unless otherwise noted.

Moving from column 1 to column 2, the coefficients shrink toward zero as the controls absorb demographic composition, except for the body-shape index, which stays positive and significant throughout. Adding the control function (columns 3–5) then moves the estimates in the direction the selection evidence of Section 4.1 predicts: wherever the selection term δ is negative, healthier types eat more FAFH, so the naive slope understates the harm and the corrected coefficient sits above the uncorrected one. For the female metabolic index, the naive no-controls estimate of 1,000 FAFH calories is -0.067 standard deviations, while the tail-symmetric estimate is 0.074 . For females, an additional 1,000 FAFH calories raises the metabolic index by 0.074 standard deviations, the body-shape index by 0.169 , and the cardiovascular index by 0.054 (the last significant at the ten percent level). Adjusted SRH does not respond for either gender. Across the corrected columns, the estimated selection terms shrink in magnitude from the uniform to the tail-symmetric specification in the rows with substantial selection (the female metabolic and cardiovascular indexes and the HEI for both genders), consistent with tail symmetry being the conservative construction (Section 4.2); rows with selection terms near zero move little across specifications.

The HEI splits into two channels of comparable size. The naive with-controls coefficient is large and negative (a loss of about 4.1 HEI points per 1,000 FAFH kcal for females and 2.2 for males), which reads as FAFH consumers eating a substantially worse diet. The bunching design attributes roughly half of that association to selection for females and a third for males: the causal effect is -1.95 HEI points per 1,000 kcal for females and -1.34 for males (both significant at the five percent level), while the selection terms are -1.74 and -0.68 . Eating out therefore does lower diet quality at the margin, and the people who eat out most would also eat lower-HEI diets regardless. Both channels that naive OLS conflates are active here, and the design separates them; for males the HEI is the only outcome with a detected causal effect, so the male harm from FAFH runs through diet composition without yet registering in the physiological indexes.

The estimates are stable across the distributional specifications within Table 7 (columns 3–5), and Section 6.1 examines the sensitivity in detail.

5.2 Heterogeneity by Age

Splitting the sample by age group reveals a second dimension of heterogeneity, shown in Figures 12a (females) and 12b (males). Females show broadly similar effects across age groups, with the 65+ estimates noisier due to the smaller sample and the metabolic point estimate closer to zero in that group; the body-shape estimates are the most stable across ages.

The coefficients, expressed per 1,000 FAFH kcal, are similar across age groups, but exposure is

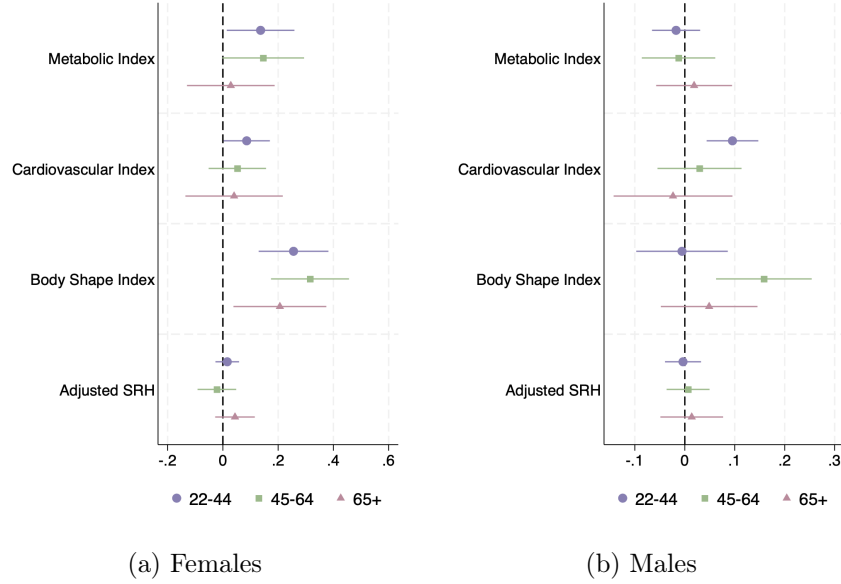


Figure 12: Tail-symmetric estimates of the effect of an additional 1,000 FAFH kcal/day on each health index, separately by gender and age group.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the results from a regression of metabolic, cardiovascular, body indexes, and adjusted SRH on the amount of calories consumed in FAFH, using as controls the demographic characteristics and clusters, separating each sample by gender and age groups.

Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

not. From Table 5, younger females consume about 40% of their calories from FAFH while those aged 65 and older consume about 30%. With a stable per-kcal effect and higher intake, the total health burden of FAFH is therefore largest for younger females, even though no age group is more sensitive than another at the margin.

Males show more heterogeneity across age groups (Figure 12b). Males aged 45–64 show the largest rise in the body-shape index (about 0.16 per 1,000 kcal) and the 22–44 group the largest rise in the cardiovascular index (about 0.10), both significant. The 65+ group has no statistically significant effect on any index, and its point estimates are small and mixed in sign. Table 6 shows the same FAFH-share pattern (about 43% for younger males, 30% for the 65+ group). A plausible mechanism is that younger males eat a worse diet whose maintenance into middle age produces weight gain. The insignificant older-male results could reflect existing medical treatment masking FAFH effects, though a parallel medication analysis found no significant effects.

To further compare the age groups, and even the genders, on a common scale, the treatment can be re-expressed as the share of daily calories coming from FAFH, so that differences in total intake across groups no longer drive the comparison.

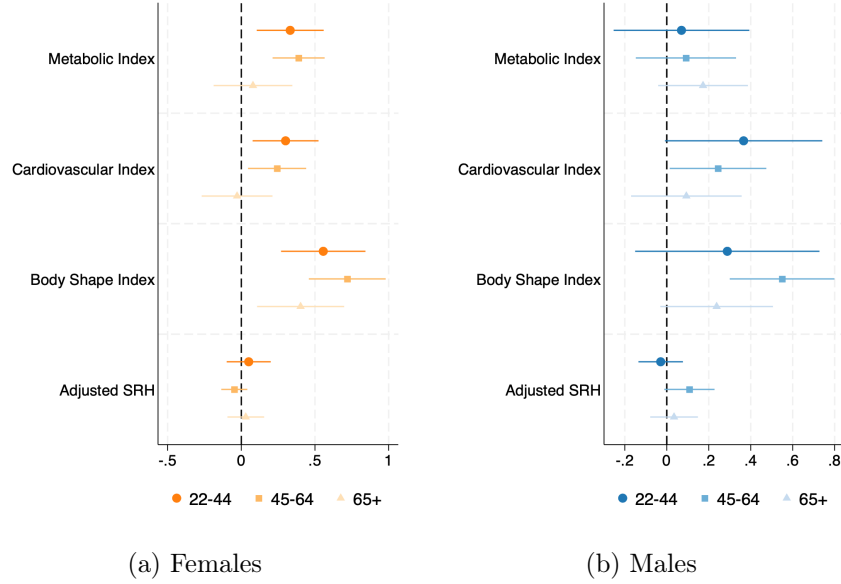


Figure 13: Tail-symmetric estimates by gender and age group when the treatment is re-expressed as the *proportion* of daily calories coming from FAFH instead of FAFH calorie counts.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the results from a regression of metabolic, cardiovascular, body indexes, and adjusted SRH on the *proportion* of total daily calories that come from FAFH (rather than on FAFH calories directly), using as controls the demographic characteristics and clusters, separating each sample by gender and age groups. The coefficient is therefore the effect of moving from a diet with 0% FAFH calories to one with 100% FAFH calories, holding daily total intake fixed; multiplying by $0.01 \cdot \overline{\text{total daily kcal}}$ in the age $\times$ gender cell of interest converts the proportion-based coefficient back to a per-additional-FAFH-kcal scale comparable to Figure 12. Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

5.3 Mechanisms

The headline estimates in Section 5.1 document the per-1,000-kcal effect of FAFH on each health index. This subsection unpacks two mechanisms that operate through diet composition rather than the indexes themselves, namely which macronutrients a marginal FAFH kcal carries (Section 5.3.1) and whether that kcal adds to the day's total intake or substitutes for FAH consumption over the same day (Section 5.3.2). Reading these alongside the headline estimates helps interpret the gender pattern, and Section 5.3.2 closes by mapping the mechanisms onto it.

5.3.1 Macronutrient Composition

FAFH meals are larger and more calorie-dense than home-prepared meals (Binkley, 2008; Prentice & Jebb, 2003), but their nutritional composition matters too. The estimates below use the same 1,000-kcal treatment scale as the health-index results.

Total caloric intake drives body-weight changes, and FAFH consumption is more calorie-dense than FAH, but the composition of those calories matters for long-run health since different macronutrients have distinct recommended intake levels.

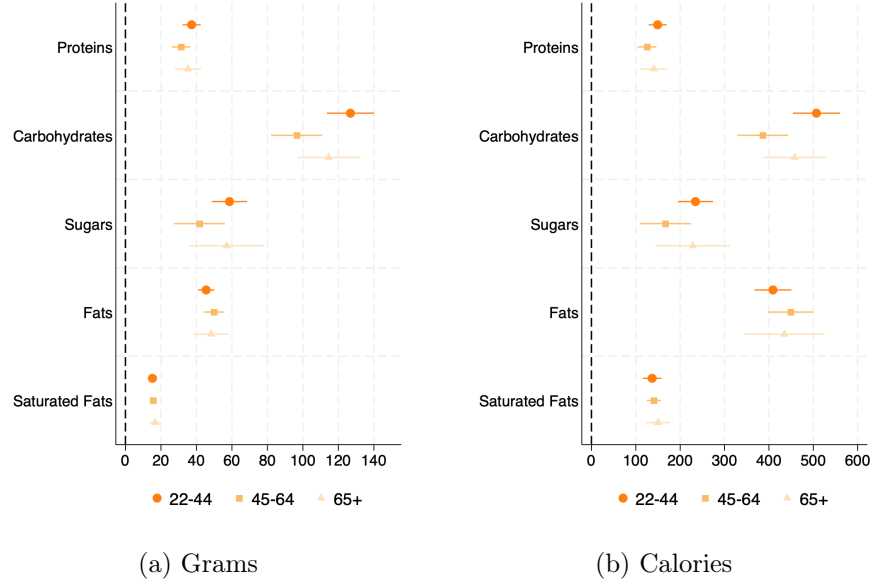


Figure 14: Effect of an additional 1,000 FAFH calories on macronutrient intake for the female sample, reported both in grams and in calorie contribution, by age group.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the results from a regression of each macronutrient, sugar and saturated fat on the amount of calories consumed in FAFH, using as controls the demographic characteristics and clusters, separating the female sample by age groups.

Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

Figure 14a shows that an additional 1,000 FAFH calories raise female carbohydrate intake by roughly 100–130 grams depending on age group, fats by 45–50 grams, and proteins by 30–37 grams. In caloric contribution (Figure 14b), carbohydrates account for the largest increase (roughly 385–505 calories), but fats add nearly as much (about 410–450 calories) despite the smaller weight gain, because fat is 9 calories per gram against 4 per gram for carbohydrates and protein (so 1,000 calories from fat weigh about 110 grams while 1,000 from carbohydrates or protein weigh about 250). Protein intake rises by roughly 125–150 calories.

Males show the same broad pattern (Figure 15a), with one exception by age: carbohydrates rise by about 130 grams per 1,000 FAFH calories for ages 22–44 and 110 for 45–64, but only about 80 grams for the 65+ group, whose carbohydrate contribution (roughly 315 calories) sits well below the 440–530 of the younger groups (Figure 15b). Fats contribute roughly 430–500 calories across ages, again delivering substantial caloric intake despite their lower weight.

Comparing the macronutrient response to 1,000 FAH calories against 1,000 FAFH calories previews the substitution channel. Since a marginal calorie carries a calorie of macronutrients from either source by construction, the differences in Figure 16 reflect how total daily intake responds to each source: most differences are positive, so a marginal FAH calorie comes with a larger increase in the day’s total macronutrient intake than a marginal FAFH calorie, meaning FAH calories are less offset by reduced consumption elsewhere in the day. The exception is males aged 22–44, for

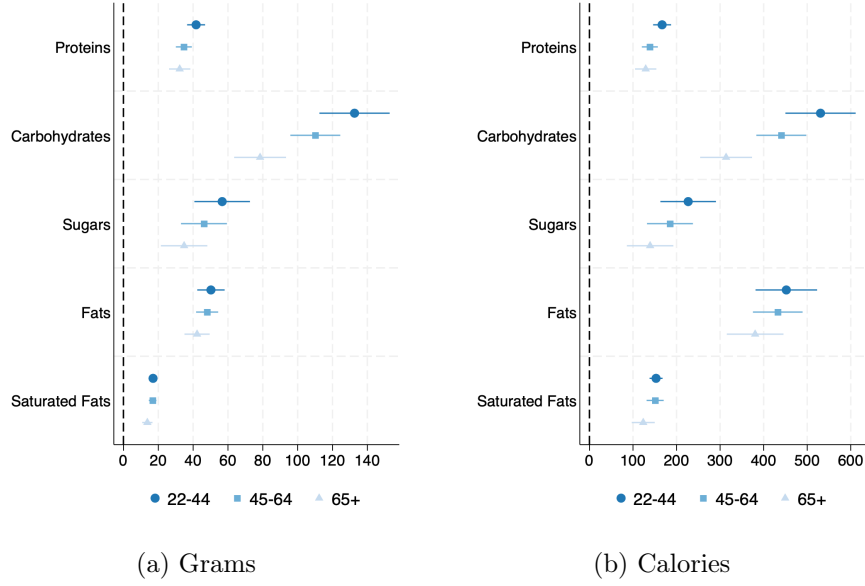


Figure 15: Effect of an additional 1,000 FAFH calories on macronutrient intake for the male sample, reported both in grams and in calorie contribution, by age group.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the results from a regression of each macronutrient, sugar and saturated fat on the amount of calories consumed in FAFH, using as controls the demographic characteristics and clusters, separating the male sample by age groups.

Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

whom the differences are indistinguishable from zero, so FAH and FAFH calories move their daily totals identically. Section 5.3.2 estimates the substitution margin directly.

5.3.2 Substitution between FAFH and FAH

Figure 17 reports the slope b from the regression

$$C_i^{\text{FAH}} = a + b C_i^{\text{FAFH}} + X_i' \gamma + e_i,$$

estimated with the control function of Section 4.2 included among the regressors, on the same analytic sample and control vector X_i as the headline health-outcome regressions, so b is the selection-corrected causal slope of home intake in FAFH intake. Three benchmark values organize the interpretation. Under perfect within-day substitution, $b = -1$: every extra FAFH calorie is offset by one fewer FAH calorie and total intake is unchanged. Under $b = 0$ there is no substitution at all: FAFH calories are added on top of an unchanged FAH intake, and total intake rises one-for-one. Positive values indicate complementarity, with total intake rising by more than the FAFH calorie itself. In general, a marginal FAFH calorie raises total daily intake by $1 + b$ calories.

For females, the coefficient is close to zero at ages 45–64 and modestly positive at 22–44 (about +0.12) and 65+ (about +0.03): no substitution to mild complementarity, so a marginal FAFH calorie adds to the day’s intake roughly one-for-one. Males differ sharply by age. The youngest

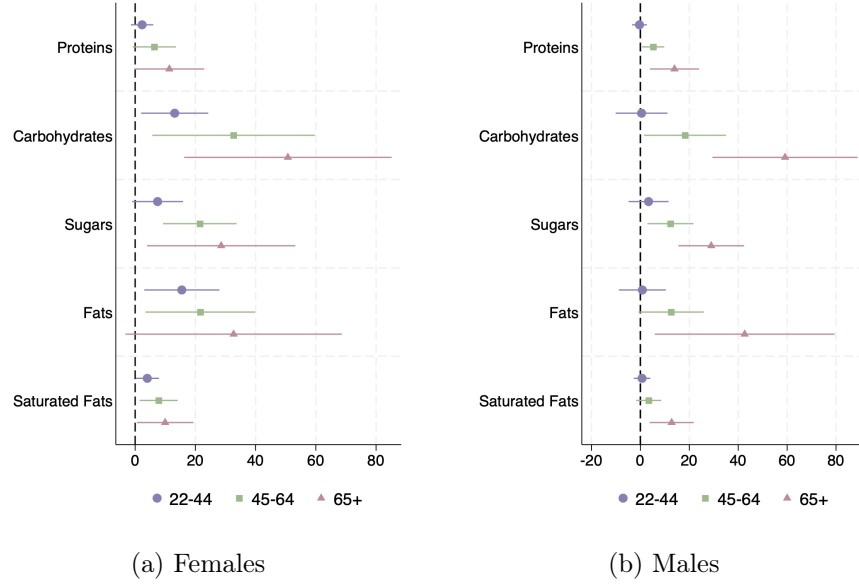


Figure 16: Difference in macronutrient intake response to an additional 1,000 calories from FAH versus FAFH, by gender and age group.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure represents the difference on results from a regression of each macronutrient, sugar and saturated fat on the amount of calories consumed in FAH and FAFH, using as controls the demographic characteristics and clusters.

Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

group shows clear complementarity (about +0.26, so total intake rises by roughly 1.26 calories per FAFH calorie), the 45–64 group sits in between (about +0.10), and only the 65+ group substitutes, and partially at that (about −0.15, so roughly 15% of a marginal FAFH calorie is offset by eating less at home).

These margins help interpret the headline estimates, though they do not explain them alone. For females, FAFH calories land on top of home intake, consistent with the body-shape and metabolic responses in Section 5.1. But young males add FAFH calories on top even more strongly than females while showing no body-shape response, so the female physiological effects must reflect more than caloric arithmetic, pointing to the composition channel of Section 5.3.1 and to physiological differences in how the same marginal intake is absorbed.

6 Robustness Checks

6.1 Distribution Assumption

Tail symmetry (Section 4.2) is the preferred assumption because it is the least parametric and the most conservative. Results can also be presented with no distributional assumption at all.

Figures 18 (females) and B.4 (males) display $\hat{\beta}$ across values of $\mathbb{E}[C^*|C = 0]$ from −1 to −1,000 for each index, with the three main distributional anchors marked as vertical lines.

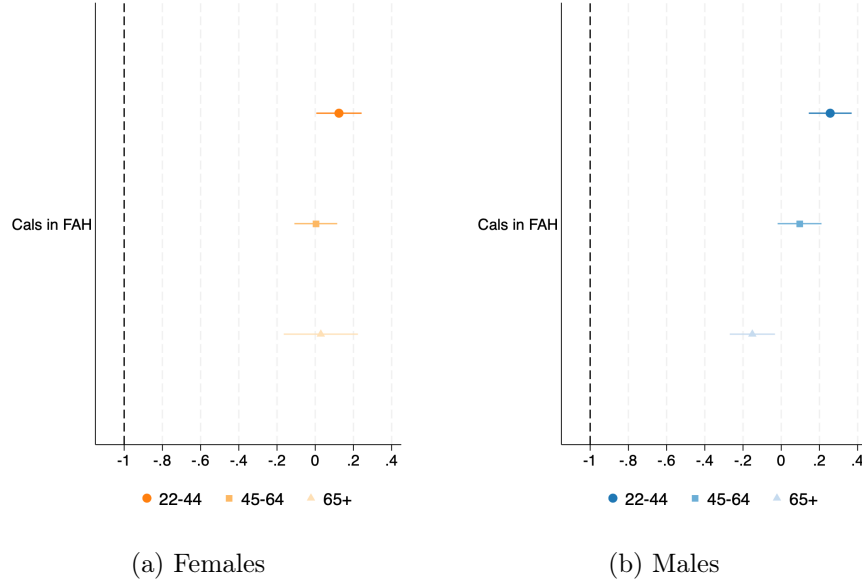


Figure 17: Bunching-corrected estimate of the slope of FAH calories on FAFH calories, by gender and age group, measuring within-day caloric substitution between the two sources.

The sweep shows that the sign of the effect does not hinge on the distributional choice: an additional 1,000 FAFH calories worsens the female health indexes across the displayed range of $\mathbb{E}[C^*|C = 0]$, and the estimates only strengthen as the assumed latent gap narrows, so the tail-symmetric headline value is the conservative end of the range.

Substitution rate on the kcal margin. Rerunning the distribution-sensitivity sweep with total daily calories (`avg_cals_0` + `avg_cals_1`) as the outcome makes $\hat{\beta}$ the substitution-rate parameter, showing how many of an additional 1,000 FAFH kcal add to daily intake versus replace home-prepared calories. Figure 19 shows the result for both genders. Under the preferred tail-symmetric anchor the slope sits near the upper end of the displayed range, so most additional FAFH calories add to total intake rather than substituting from home food. The qualitative reading is stable across all three distributional assumptions, so the substitution-channel interpretation in Section 5 does not depend on the distribution.

6.2 Linearity of Extrapolation

The identification extrapolates the positive-side relationship between outcomes and consumption linearly to the corner, with the level function $m(X)$ piecewise constant over the $K = 20$ clusters of X . The estimates could therefore in principle be driven by observations far from the bunching point. The standard diagnostic is to re-estimate $\hat{\beta}$ on progressively narrower sub-samples defined by $C \leq H_{\max}$: stability as $H_{\max}$ shrinks indicates that the far tail is not doing the work.

Figures 20 and 21 show $\hat{\beta}$ across $H_{\max} \in \{500, 750, 1000, 1500, 2000, 2500, 3000\}$ kcal/day for each outcome and gender. The widest bandwidth retains essentially the full sample (the 99th percentile of `avg_cals_1` is 2,775 kcal/day). Two patterns emerge. From $H_{\max} = 1,500$, which still

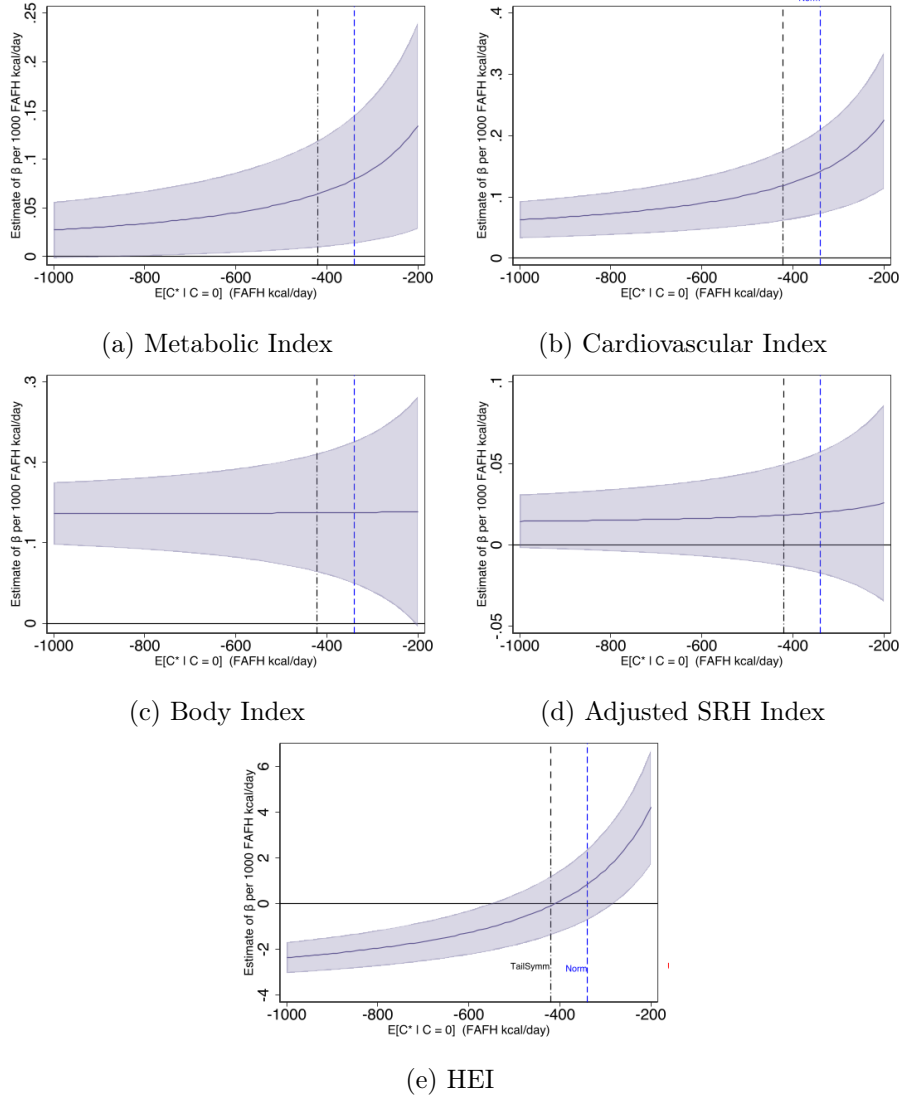


Figure 18: Distribution-sensitivity check for the female sample: $\hat{\beta}$ for each outcome as the assumed counterfactual mean $\mathbb{E}[C^* | C = 0]$ is varied across the support implied by the three reference distributional assumptions.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Each Figure shows the values of $\hat{\beta}$ for different values of $\mathbb{E}[C^* | C = 0]$, using the female sample. The vertical lines black, blue, and red represent the values for the expectation for semiparametric uniform, semiparametric normal, and nonparametric tail symmetric distributions, respectively.

Bootstrapped confidence intervals of 95% are shown on the plot, based on 100 iterations.

keeps roughly 93% of the sample, point estimates are stable and close to the main-table values for every outcome and both genders, so the far tail is not driving the headline results. Below 1,500 the point estimates rise as the window narrows, most visibly for the body-shape index (to roughly 0.5 for females and 0.3 for males at $H_{\max} = 500$, against current headline values of 0.169 and 0.016) and the female metabolic index (to roughly 0.25), with confidence intervals wide enough that the narrow-window estimates remain compatible with the headline values. This rise is the imprint of

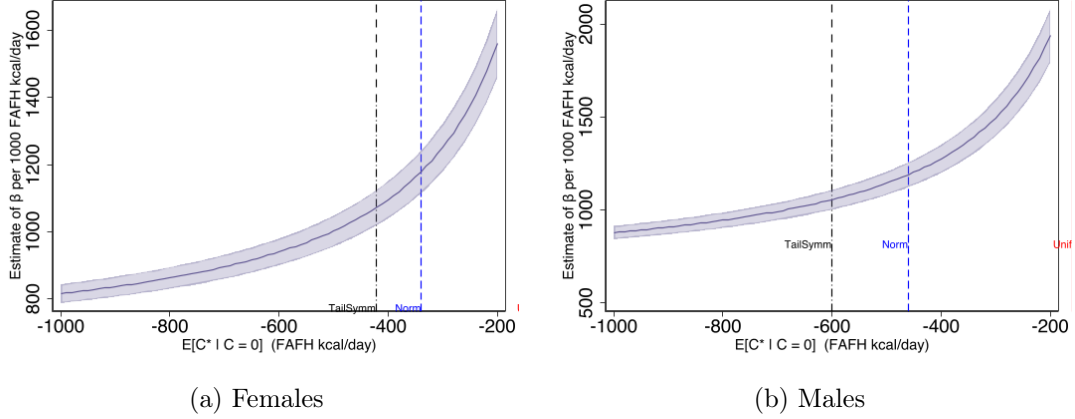


Figure 19: Distribution-sensitivity sweep applied to the FAFH-to-total-calories substitution rate, by gender, testing whether the substitution-channel estimate survives the bunching distributional assumption.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: Same distribution-sensitivity sweep as Figures 18 and B.4, but with the outcome replaced by total daily calories (`avg_cals_0 + avg_cals_1`). The slope $\hat{\beta}$ is therefore the change in total daily calories per additional 1,000 FAFH kcal: a value near 1,000 means full caloric addition (no substitution from home food), and a value near 0 means full substitution. Vertical lines mark the implied $E[C^* | C = 0]$ under the semiparametric uniform (red), semi-parametric normal (blue, dashed), and nonparametric tail symmetric (black, dash-dot) assumptions. Bootstrapped 95% confidence intervals based on 100 iterations.

the curvature visible in the residualized profiles of Section 4.1: near the corner, the body-shape and metabolic profiles are steeper than their full-range slope, so estimation windows confined to the corner deliver larger effects. The headline estimates are accordingly read as effects identified over the full range of observed consumption under the linear specification, and the near-corner windows as suggestive of larger local responses that the data are too thin to pin down precisely.

6.3 Number of Clusters

Two diagnostics for the clustering step follow. Figure 22 shows how FAFH calories are distributed within and across the $K = 20$ clusters of X used in the main estimation. Figures 23 and B.5 sweep the number of clusters from $K = 1$ to $K = 30$ and show that the coefficient is stable across this range for both genders.

Figure 22 addresses the concern that the bunching identification could be driven by the clustering of X rather than by genuine variation in FAFH consumption. For each of the $K = 20$ clusters, the within-cluster median of `avg_cals_1` is plotted against the rank of that cluster in the sorted median, with interquartile whiskers on four spaced-out clusters. The medians span roughly 200 to 700 kcal/day for both genders, and the within-cluster IQRs are of comparable magnitude. Treatment heterogeneity therefore exists both across clusters (the gradient that $m(X)$ absorbs) and within clusters (the residual variation that identifies β at the bunching margin). This is the empirical analog of CCNS Maternal Section 3.4.2 and supports the use of the cluster control as a valid

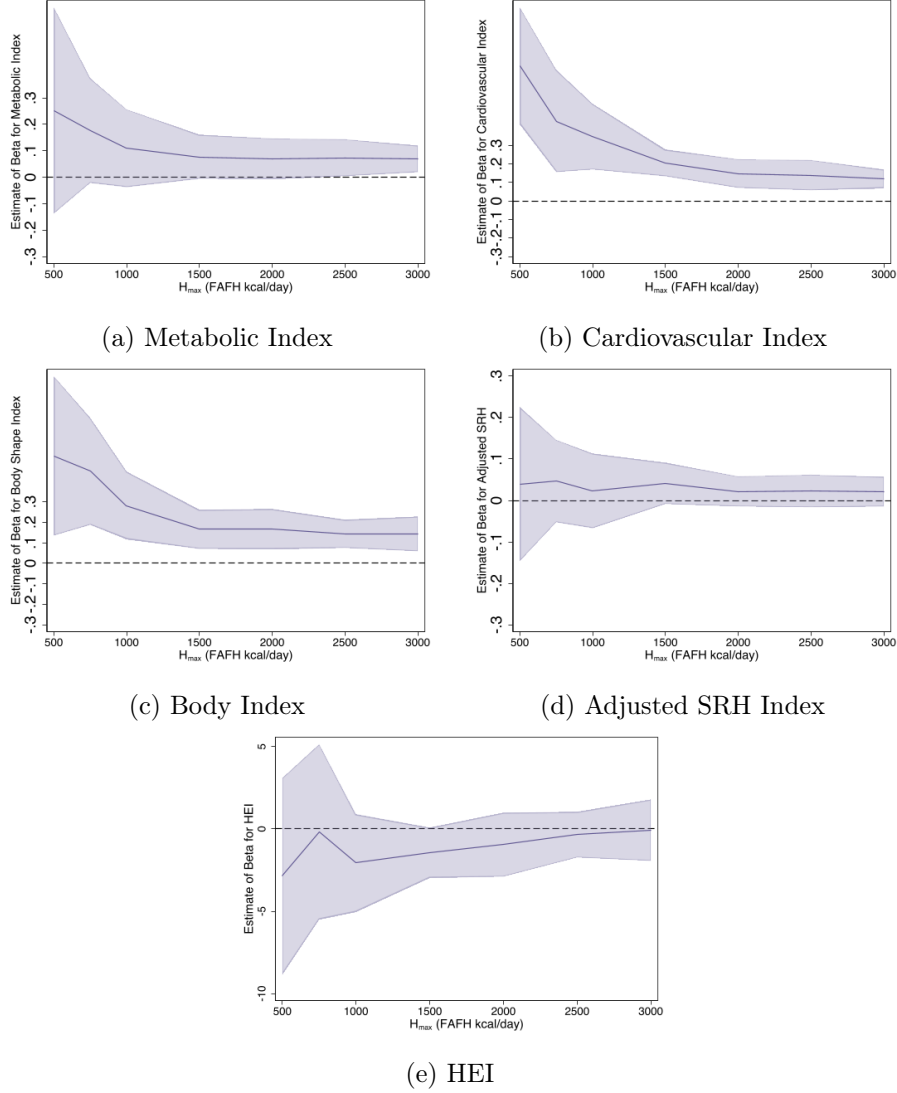


Figure 20: Linearity-of-extrapolation robustness check for the female sample: $\hat{\beta}$ re-estimated on the sub-sample $\text{avg_cals_1} \leq H_{\max}$, varying $H_{\max}$ from 500 to 3,000 kcal/day.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: The figure shows the values of $\hat{\beta}$ (per 1,000 FAFH kcal/day) re-estimated on the sub-sample defined by $\text{avg_cals_1} \leq H_{\max}$, varying $H_{\max}$ from 500 to 3,000 kcal/day. A flat profile means the control-function extrapolation is not being driven by far-from-bunching observations.

Bootstrapped 95% confidence intervals based on 25 iterations. Distributional assumption is the nonparametric tail symmetric distribution.

analogue for the conditional expectation of X .

Figure 23 presents estimates of β for $K = 1$ to $K = 30$ for the female sample. The point estimates barely move across the range. Males (Figure B.5) show the same pattern.

The main analysis uses 20 clusters, chosen from the cluster-sweep results (particularly Figure B.5c) where estimates stabilize around $K = 20$.

Beyond validating the cluster count, this analysis supports Assumption 1, since the clusters

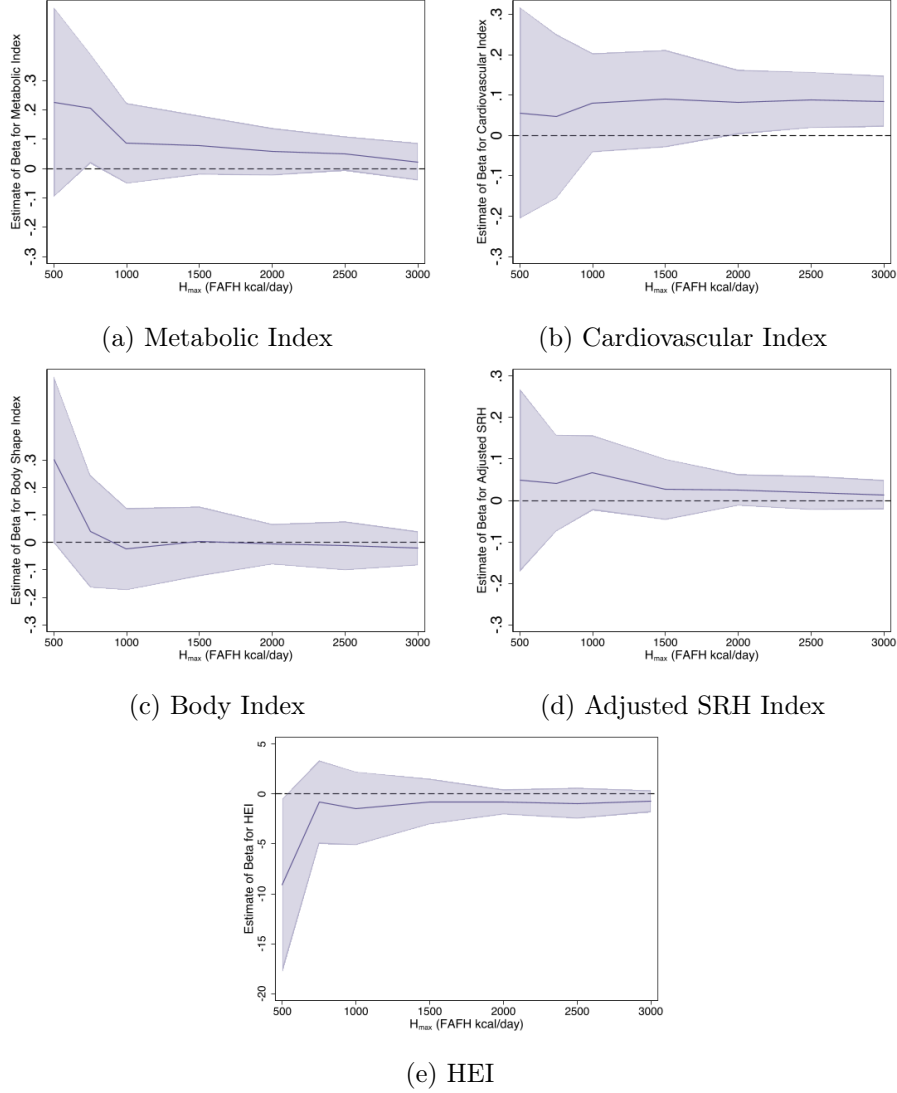


Figure 21: Linearity-of-extrapolation robustness check for the male sample: $\hat{\beta}$ re-estimated on the sub-sample $\text{avg_cals_1} \leq H_{\max}$, varying $H_{\max}$ from 500 to 3,000 kcal/day.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: The figure shows the values of $\hat{\beta}$ (per 1,000 FAFH kcal/day) re-estimated on the sub-sample defined by $\text{avg_cals_1} \leq H_{\max}$, varying $H_{\max}$ from 500 to 3,000 kcal/day. A flat profile means the control-function extrapolation is not being driven by far-from-bunching observations.

Bootstrapped 95% confidence intervals based on 25 iterations. Distributional assumption is the nonparametric tail symmetric distribution.

capture a substantial share of the variation in the choice of C .

6.4 Treatment Variable

The headline treatment is FAFH calories from the first 24-hour recall. That choice follows the NIH guidance that a single recall suffices for population-level analysis (Section 3.1), but it is still a choice, so this section re-estimates the model under alternative definitions of the treatment: the two-recall

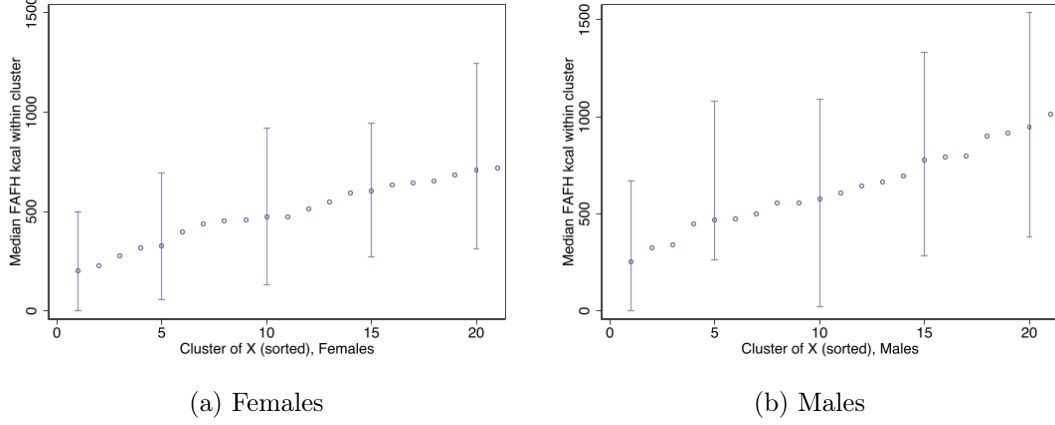


Figure 22: Distribution of FAFH calories within and across the $K = 20$ clusters of X used in the main estimation, separately by gender.

Data source: National Health and Nutrition Examination Survey (NHANES).

Notes: For each of the $K = 20$ clusters of X , the within-cluster median of `avg_cals_1` (daily FAFH calories) is plotted against the cluster’s rank in the sorted median. Vertical whiskers show the interquartile range of FAFH calories within four spaced-out clusters (cluster ranks 1, $K/4$, $K/2$, $3K/4$, K). The pattern across clusters shows that FAFH intake varies substantively both *across* clusters (a roughly monotone gradient from ~ 200 to ~ 700 kcal) and *within* clusters (IQRs of several hundred kcal), so the bunching identification rests on real, not synthetic, heterogeneity in the treatment variable.

average, and the second recall alone.

The two-recall average is constructed to respect typicality. For respondents who completed only one interview, intake comes from that day, provided they reported it as a typical eating day. For respondents who completed both, intake from the typical day is used, or the average of the two when both were typical; individuals with no typical eating day are excluded. Averaging smooths day-to-day reporting variation, at two costs: it shrinks the bunching mass that identifies the design, since a respondent must report zero FAFH on *both* days to remain at the corner (the zero share falls from roughly 31% to 18%), and it conditions in part on the voluntarily completed second interview.

Figure 24 shows the tail-symmetric estimates under the average definition. The qualitative pattern of the main results is preserved: adverse effects concentrate among females, and adjusted SRH is a null for both genders. The magnitudes shift in instructive ways, with a larger cardiovascular estimate and a somewhat smaller body-shape estimate than under the day-1 definition, and the diet-quality effect is where the definition matters most, a point the discussion in Section 5 returns to.

7 Conclusion

This study examines the impact of Food Away from Home (FAFH) consumption on health across genders, focusing on the cardiovascular index, metabolic index, body-shape index, adjusted SRH, and diet quality measured by the HEI.

Using NHANES data, the analysis applies a control-function approach that accounts for endogenous selection into eating habits. The method exploits the assumption that treatment effects are

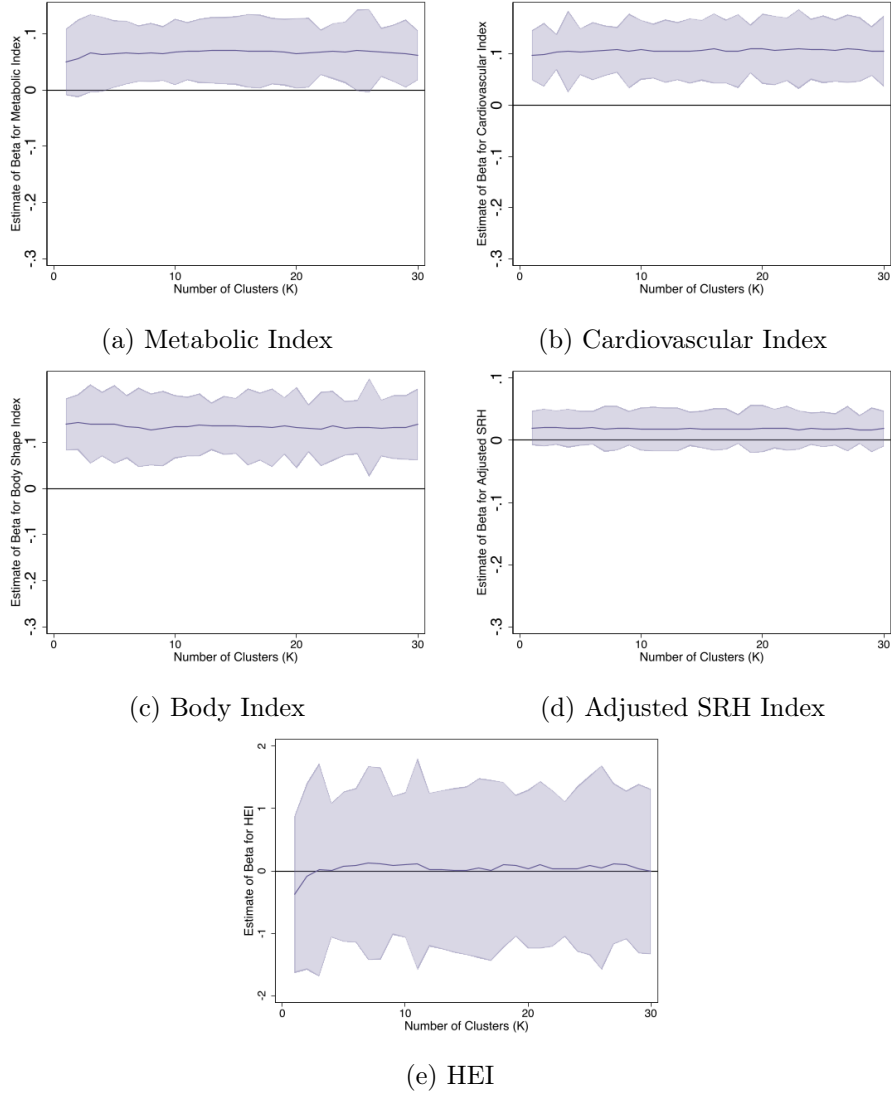


Figure 23: Robustness check for the number of clusters K in the control function $m(X)$, for the female sample: $\hat{\beta}$ re-estimated for $K = 1, \dots, 30$.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The Figure shows the values of $\hat{\beta}$ for different numbers of clusters K , for the female sample. Bootstrapped confidence intervals of 95% are shown on the plot, based on 25 iterations.

Distributional assumption is the nonparametric tail symmetric distribution.

continuous while confounders show discontinuities at zero FAFH calories due to bunching, using outcome-side discontinuities to identify the confounder effects. After the endogeneity correction, FAFH consumption is associated with adverse health outcomes, more strongly for females than males.

An additional 1,000 FAFH calories significantly worsens the metabolic, cardiovascular, and body-shape indexes for females, and lowers diet quality (HEI) for both genders. For males, no physiological index responds significantly; the male effect runs through diet quality. Females show uniform effects

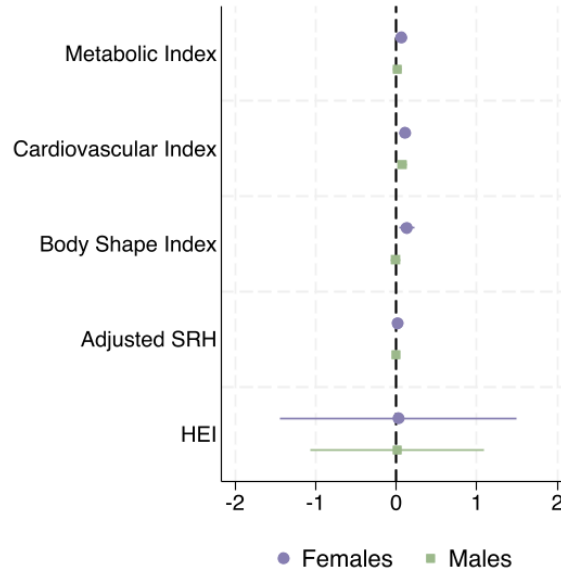


Figure 24: Robustness check for the treatment variable: re-estimation using the typicality-adjusted average of both 24-hour dietary recalls instead of the first recall alone.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The figure reports the tail-symmetric estimates of the effect of an additional 1,000 FAFH kcal/day on the metabolic, cardiovascular, and body-shape indexes and adjusted SRH, using the demographic controls and clusters, separately by gender.

Bootstrapped confidence intervals of 95% are provided based on 250 iterations.

across ages, while the male age profile is noisier.

The contribution is to move beyond the established link between FAFH and obesity by incorporating metabolic and cardiovascular indexes that together give a more complete picture of how dietary choices influence health.

The magnitudes align with the two closest causal-design papers in the FAFH literature while isolating channels neither can. Mancino et al. (2009) use a first-difference estimator on dietary intakes and find each FAFH meal adds roughly 134 kcal and reduces daily diet-quality scores. The bunching-corrected estimates translate to a comparable nutritional shift and further show that the impact propagates to multi-component biomarker indexes with magnitudes of 0.05–0.17 standard deviations per 1,000 FAFH kcal for females. Currie et al. (2010) measure the effect of proximity to a fast-food restaurant and report obesity rates roughly 5% higher for ninth-graders within 0.1 mile of one, aligning in sign with the positive female body-shape coefficient documented here, though identified off school-level proximity, which cannot separate the metabolic and cardiovascular channels that the bunching design recovers.

NHANES’s self-reported dietary intake may not perfectly reflect actual FAFH consumption, which limits the interpretation.

Future work could examine the mechanisms behind the FAFH effects, the differential impact of fast food, full-service restaurants, and vending machines, and the mediating role of physical activity.

This study documents the detrimental effects of FAFH consumption on health, especially for women, and offers a methodological framework for analyzing dietary behavior and health.

References

- Abrahamsson, S., Butikofer, A., & Karbownik, K. (2023). Swallow this: Childhood and adolescent exposure to fast food restaurants, bmi, and cognitive ability. *SSRN Electronic Journal*.
- Alviola, P. A. I., Nayga, R. M. J., Thomsen, M. R., Danforth, D., & Smartt, J. (2014). The effect of fast-food restaurants on childhood obesity: A school level analysis. *Economics and Human Biology*, 12, 110–119.
- American Heart Association. (2025). Fast heart rate [Accessed: 30 Mar. 2025]. <https://www.heart.org/en/health-topics/arrhythmia/about-arrhythmia/tachycardia--fast-heart-rate>
- Austin, M. A., Hokanson, J. E., & Edwards, K. L. (1998). Hypertriglyceridemia as a cardiovascular risk factor. *The American Journal of Cardiology*, 81(4A), 7B–12B. Retrieved April 29, 2024, from <https://www.bls.gov/spotlight/2020/food-away-from-home/home.htm>
- Bania, N., & Leete, L. (2022). Monthly income volatility and health outcomes. *Contemporary Economic Policy*, 40(4), 636–658.
- Barnard, R. J. (1991). Effects of life-style modification on serum lipids. *Archives of Internal Medicine*, 151(7), 1389–1394.
- Bertoli, S., Leone, A., Krakauer, N. Y., Bedogni, G., Vanzulli, A., Redaelli, V. I., De Amicis, R., Vignati, L., Krakauer, J. C., & Battezzati, A. (2017). Association of body shape index (absi) with cardio-metabolic risk factors: A cross-sectional study of 6081 caucasian adults. *PLOS ONE*, 12(9), e0185013.
- Binkley, J. K. (2008). Calorie and gram differences between meals at fast food and table service restaurants. *Review of Agricultural Economics*, 30(4), 750–763.
- Binkley, J. K., Eales, J., & Jekanowski, M. (2000). The relation between dietary change and rising us obesity. *International Journal of Obesity*, 24(8), 1032–1039.
- Boeing, H., et al. (2000). Association between glycated hemoglobin and diet and other lifestyle factors in a nondiabetic population: Cross-sectional evaluation of data from the potsdam cohort of the european prospective investigation into cancer and nutrition study. *The American Journal of Clinical Nutrition*, 71(5), 1115–1122. <https://doi.org/10.1257/pol.6.2.258>
- Bombak, A. E. (2013). Self-rated health and public health: A critical perspective. *Frontiers in Public Health*, 1, 15.
- Bound, J. (1991). Self-reported versus objective measures of health in retirement models. *The Journal of Human Resources*, 26(1), 106–138.
- Bowman, S. A., & Vinyard, B. T. (2004). Fast food consumption of u.s. adults: Impact on energy and nutrient intakes and overweight status. *Journal of the American College of Nutrition*, 23(2), 163–168.
- Bureau of Labor Statistics. (2020). *Meal appeal: Patterns of expenditures on food away from home*.
- Caetano, C., Caetano, G., Mazetto, D., & Skira, M. (2025). *The effect of video watching on children’s cognitive skills* [Working paper].
- Caetano, C., Caetano, G., & Nielsen, E. (2024a). Correcting for endogeneity in models with bunching. *Journal of Business & Economic Statistics*, 42(3), 851–863.
- Caetano, C., Caetano, G., Nielsen, E., & Sanfelice, V. (2024b). *The effect of maternal labor supply on children: Evidence from bunching* (Technical report). Forthcoming, Journal of Labor Economics.
- Caetano, C., Caetano, G., & Nielsen, E. R. (2024c). *Are children spending too much time on enrichment activities?* [Working paper].
- Caetano, C., Caetano, G., & Santos, K. (2026). *Household chores and the gender gap in child development: Evidence from bunching* [Working paper].

- Chou, S.-Y., Grossman, M., & Saffer, H. (2004). An economic analysis of adult obesity: Results from the behavioral risk factor surveillance system. *Journal of Health Economics*, 23(3), 565–587.
- Currie, J., DellaVigna, S., Moretti, E., & Pathania, V. (2010). The effect of fast food restaurants on obesity and weight gain. *American Economic Journal: Economic Policy*, 2(3), 32–63.
- Cutler, D. M., Glaeser, E. L., & Shapiro, J. M. (2003). Why have americans become more obese? *Journal of Economic Perspectives*, 17(3), 93–118.
- Duffey, K. J., Gordon-Larsen, P., Jacobs Jr, D. R., Williams, O. D., & Popkin, B. M. (2007). Differential associations of fast food and restaurant food consumption with 3-y change in body mass index: The coronary artery risk development in young adults study2. *The American Journal of Clinical Nutrition*, 85(1), 201–208.
- Evans, W. N., & Garthwaite, C. L. (2014). Giving mom a break: The impact of higher eite payments on maternal health. *American Economic Journal: Economic Policy*, 6(2), 258–90.
- Franks, P., Gold, M. R., & Fiscella, K. (2003). Sociodemographics, self-rated health, and mortality in the us. *Social Science & Medicine*, 56(12), 2505–2514.
- Gillum, R. F., Makuc, D. M., & Feldman, J. J. (1991). Pulse rate, coronary heart disease, and death: The nhanes i epidemiologic follow-up study. *American Heart Journal*, 121(1), 172–177.
- Goffe, L., Rushton, S., White, M., et al. (2017). Relationship between mean daily energy intake and frequency of consumption of out-of-home meals in the uk national diet and nutrition survey. *International Journal of Behavioral Nutrition and Physical Activity*, 14, 131.
- Hoynes, H., Schanzenbach, D. W., & Almond, D. (2016). Long-run impacts of childhood access to the safety net. *American Economic Review*, 106(4), 903–34.
- Idler, E. L., & Angel, R. J. (1990). Self-rated health and mortality in the nhanes-i epidemiologic follow-up study. *American Journal of Public Health*, 80(4), 446–452.
- Johns Hopkins Medicine. (2025). Lipid panel [Accessed: 30 Mar. 2025]. <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/lipid-panel>
- Kant, A. K., Whitley, M. I., & Graubard, B. I. (2015). Away from home meals: Associations with biomarkers of chronic disease and dietary intake in American adults, NHANES 2005-2010. *International Journal of Obesity*, 39(5), 820–827.
- Kleven, H. J. (2016). Bunching. *Annual Review of Economics*, 8, 435–464.
- Kling, J. R., Liebman, J. B., & Katz, L. F. (2007). Experimental analysis of neighborhood effects. *Econometrica*, 75(1), 83–119.
- Krakauer, N. Y., & Krakauer, J. C. (2012). A new body shape index predicts mortality hazard independently of body mass index. *PLOS ONE*, 7(7), e39504.
- Larson, N., Neumark-Sztainer, D., Laska, M. N., & Story, M. (2011). Young adults and eating away from home: Associations with dietary intake patterns and weight status differ by choice of restaurant [Available at SSRN: <https://ssrn.com/abstract=1557129> or <http://dx.doi.org/10.2139/ssrn.1557129>]. *Journal of the American Dietetic Association*, 111(11), 1696–1703. <https://ssrn.com/abstract=1557129>
- Lin, B.-H., & Guthrie, J. (2012). *Nutritional quality of food prepared at home and away from home, 1977-2008, eib-105*. U.S. Department of Agriculture, Economic Research Service.
- Liu, Y., Ozodiegwu, I. D., Nickel, J. C., Wang, K., & Iwasaki, L. R. (2017). Self-reported health and behavioral factors are associated with metabolic syndrome in americans aged 40 and over. *Preventive Medicine Reports*, 7, 193–197.
- Mancino, L., Todd, J., & Lin, B.-H. (2009). Separating what we eat from where: Measuring the effect of food away from home on diet quality. *Food Policy*, 34(6), 557–562.
- Mensink, R. P., Zock, P. L., Kester, A. D., & Katan, M. B. (2003). Effects of dietary fatty acids and carbohydrates on the ratio of serum total to hdl cholesterol and on serum lipids and

- apolipoproteins: A meta-analysis of 60 controlled trials. *The American Journal of Clinical Nutrition*, 77(5), 1146–1155.
- Michaud, P.-C., Crimmins, E. M., & Hurd, M. D. (2016). The effect of job loss on health: Evidence from biomarkers. *Labour Economics*, 41, 194–203.
- Nagao-Sato, S., & Reicks, M. (2022). Food away from home frequency, diet quality, and health: Cross-sectional analysis of nhanes data 2011–2018. *Nutrients*, 14, 3386.
- National Cancer Institute. (n.d.). Dietary assessment primer: Dietary intake data collection approaches [Accessed: November 22, 2024]. <https://dietassessmentprimer.cancer.gov/approach/intake.html>
- National Cancer Institute, Division of Cancer Control and Population Sciences. (2023). Overview of the methods & calculations for the Healthy Eating Index (HEI) [SAS code for HEI-2005, HEI-2010, and HEI-2015 component scoring; accessed via the NCI Dietary Assessment Resources portal.].
- National Center for Biotechnology Information. (2025). Ncbi bookshelf: Hemoglobin A1C [Accessed: 30 Mar. 2025]. <https://www.ncbi.nlm.nih.gov/books/NBK549816/>
- National Heart, Lung, and Blood Institute. (2025). High blood pressure [Accessed: 30-Mar-2025]. <https://www.nhlbi.nih.gov/health/high-blood-pressure>
- Pereira, P., Mark A., Kartashov, P., Alex I., Ebbeling, P., Cara B., Van Horn, P., Prof Linda, Slattery, P., Prof Martha L., Jacobs Jr, P., Prof David R., et al. (2005). Fast-food habits, weight gain, and insulin resistance (the cardia study): 15-year prospective analysis [Available at SSRN]. *The Lancet*, 365(9453), 36–42. <https://ssrn.com/abstract=4724194>
- Prentice, A., & Jebb, S. (2003). Fast foods, energy density and obesity: A possible mechanistic link. *Obesity Reviews*, 4, 187–194.
- Ricciuto, L., Fulgoni, V. L. I., Gaine, P. C., Scott, M. O., & DiFrancesco, L. (2021). Sources of added sugars intake among the u.s. population: Analysis by selected sociodemographic factors using the national health and nutrition examination survey 2011-18. *Frontiers in Nutrition*, 8, 687643.
- Saez, E. (2010). Do taxpayers bunch at kink points? *American Economic Journal: Economic Policy*, 2(3), 180–212.
- Saksena, M. J., Okrent, A. M., Anekwe, T. D., Cho, C., Dicken, C., Effland, A., Elitzak, H., Guthrie, J., Hamrick, K. S., Hyman, J., Jo, Y., Lin, B.-H., Mancino, L., McLaughlin, P. W., Rahkovsky, I., Ralston, K., Smith, T. A., Stewart, H., Todd, J., & Tuttle, C. (Eds.). (2018). *America's eating habits: Food away from home, eib-196*. U.S. Department of Agriculture, Economic Research Service.
- Savica, V., Bellinghieri, G., & Kopple, J. D. (2010). The effect of nutrition on blood pressure. *Annual Review of Nutrition*, 30, 365–401.
- Schaller, J., & Stevens, A. H. (2015). Short-run effects of job loss on health conditions, health insurance, and health care utilization. *Journal of Health Economics*, 43, 190–203.
- Seeman, T., Merkin, S. S., Crimmins, E., Koretz, B., Charette, S., & Karlamangla, A. (2008). Education, income and ethnic differences in cumulative biological risk profiles in a national sample of us adults: Nhanes iii (1988-1994). *Social Science & Medicine*, 66(1), 72–87.
- Seguin, R. A., Aggarwal, A., Vermeylen, F., & Drewnowski, A. (2016). Consumption frequency of foods away from home linked with higher body mass index and lower fruit and vegetable intake among adults: A cross-sectional study. *Journal of Environmental and Public Health*, 2016, 3074241.
- Tello, M. (2020). Better heart health in eight weeks? double down on fruits and veggies [Accessed: 30 Mar. 2025]. <https://www.health.harvard.edu/blog/better-heart-health-in-eight-weeks-double-down-on-fruits-and-veggies-2020061120163>

- Todd, J. E., Mancino, L., & Lin, B.-H. (2010). *The impact of food away from home on adult diet quality* (Economic Research Report Paper No. 90). USDA-ERS.
- Vevers, S. (2024). How long may it take to achieve type 2 diabetes remission? [Accessed: 30 Mar. 2025]. <https://www.medicalnewstoday.com/articles/how-long-does-it-take-to-reverse-type-2-diabetes>
- Ware, D., Landy, D. C., Rabil, A., Hennekens, C. H., & Hecht, E. M. (2022). Interrelationships between self reported physical health and health behaviors among healthy us adults: From the nhanes 2009–2016. *Public Health in Practice*, 4, 100277.

A Tables

Table A.1: Food Source Classification

NHANES Source	Category
Store - grocery/supermarket	Home
Restaurant with waiter/waitress	Restaurant
Restaurant fast food/pizza	Fast food
Bar/tavern/lounge	Other
Restaurant no additional information	Restaurant
Cafeteria NOT in a K-12 school	Cafeteria
Cafeteria in a K-12 school	Cafeteria
Child/Adult care center	Restaurant
Child/Adult home care	Restaurant
Soup kitchen/shelter/food pantry	Food Assistance Program
Meals on Wheels	Food Assistance Program
Community food program - other	Food Assistance Program
Community program no additional information	Food Assistance Program
Vending machine	Cafeteria
Common coffee pot or snack tray	Cafeteria
From someone else/gift	Home
Mail order purchase	Home
Residential dining facility	Restaurant
Grown or caught by you or someone you know	Home
Fish caught by you or someone you know	Home
Sport, recreation, or entertainment facility	Cafeteria
Street vendor, vending truck	Fast food
Fundraiser sales	Other
Store - convenience type	Home
Store - no additional info	Home
Other, specify	Other
Don't know	Other

The classification was done following the ERS-USDA: <https://www.ers.usda.gov/data-products/food-expenditure-series/documentation/>

Table A.2: Biomarkers for Males

Biomarker	Measurement unit	Observations		Sample Mean		Ideal range	% in risk	
		FAFH=0	FAFH> 0	FAFH=0	FAFH> 0		FAFH=0	FAFH> 0
Cardiovascular measures								
Systolic blood pressure	mmHg	3,579	18,811	127.73	122.83	<130	0.368	0.254
Diastolic blood pressure	mmHg	3,560	18,674	69.7	68.82	<90	0.061	0.046
Heart Rate	beats/min	3,763	19,731	71.65	71.43	<90	0.079	0.063
Metabolic measures								
Cholesterol Total	mg/dL	3,602	19,024	186.74	183.48	<200	0.35	0.315
Cholesterol HDL	mg/dL	3,612	19,088	48.56	48.68	>60	0.813	0.818
Glycosylated Hemoglobin	(%)	3,652	19,236	5.78	5.64	<6.5	0.118	0.084
Triglycerides	mg/dL	1,702	9,211	138.36	126.11	<150	0.309	0.244
External measures								
BMI	points	3,797	19,986	27.6	27.58	<27	0.29	0.296
Waist circumference	cm	3,691	19,636	98.39	96.65	<80	0.601	0.55

Table A.3: Biomarkers for Females

Biomarker	Measurement unit	Observations		Sample Mean		Ideal range	% in risk	
		FAFH=0	FAFH> 0	FAFH=0	FAFH> 0		FAFH=0	FAFH> 0
Cardiovascular measures								
Systolic blood pressure	mmHg	3,789	18,490	125.12	119.66	<130	0.329	0.227
Diastolic blood pressure	mmHg	3,753	18,378	67.71	67.29	<90	0.031	0.028
Heart Rate	beats/min	4,041	19,563	74.51	74.94	<90	0.093	0.092
Metabolic measures								
Cholesterol Total	mg/dL	3,865	18,810	196.68	189.05	<200	0.423	0.352
Cholesterol HDL	mg/dL	3,892	18,883	56.73	57.06	>60	0.624	0.616
Glycosylated Hemoglobin	(%)	3,953	19,127	5.78	5.59	<6.5	0.117	0.072
Triglycerides	mg/dL	1,777	9,120	125.76	108.53	<150	0.258	0.184
External measures								
BMI	points	4,114	19,875	28.61	28.48	<27	0.357	0.359
Waist circumference	cm	3,960	19,353	95.14	93.81	<80	0.807	0.759

Table A.4: Effects of Calories consumed in FAFH for Males

Dependent Variable		(1) Uncorrected No Controls	(2) Uncorrected w/ Controls	(3) Semip. Uniform	(4) Semip. Normal	(5) Nonp. Tail Symmetric
Metabolic Index (N= 14,639)	β	-0.017** (0.005)	-0.001 (0.005)	-0.011 (0.030)	-0.007 (0.021)	-0.006 (0.019)
	δ			0.010 (0.027)	0.005 (0.018)	0.004 (0.016)
Cardiovascular Index (N= 15,062)	β	-0.009* (0.005)	-0.001 (0.005)	0.023 (0.033)	0.026 (0.022)	0.029 (0.020)
	δ			-0.022 (0.030)	-0.023 (0.018)	-0.025 (0.016)
Body Shape Index (N= 15,174)	β	0.102** (0.008)	0.020** (0.007)	0.000 (0.039)	0.013 (0.028)	0.016 (0.026)
	δ			0.018 (0.035)	0.006 (0.023)	0.004 (0.021)
Adjusted SRH (N= 15,274)	β	-0.058** (0.008)	0.000 (0.003)	0.001 (0.017)	0.000 (0.012)	0.000 (0.011)
	δ			-0.001 (0.015)	0.001 (0.010)	0.000 (0.009)
HEI (N= 15,274)	β	-2.825** (0.104)	-2.157** (0.113)	-0.174 (0.714)	-1.142** (0.481)	-1.343** (0.441)
	δ			-1.795** (0.650)	-0.859** (0.407)	-0.676* (0.365)

B Figures

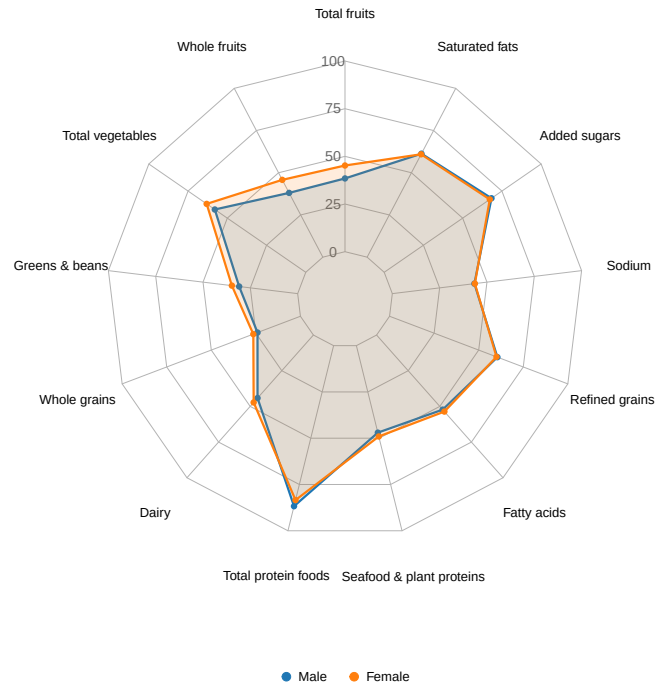


Figure B.1: HEI-2015 component scores by gender, showing the female advantage on the adequacy components and the near-equality on the moderation components.

Data source: NHANES 2011–2018. Mean of each HEI-2015 component normalized to the 0–100 scale (percent of component maximum). Higher values on every axis indicate a diet better aligned with the federal dietary guidelines (reverse-coded moderation components are oriented so that lower intake shows up as a higher score).

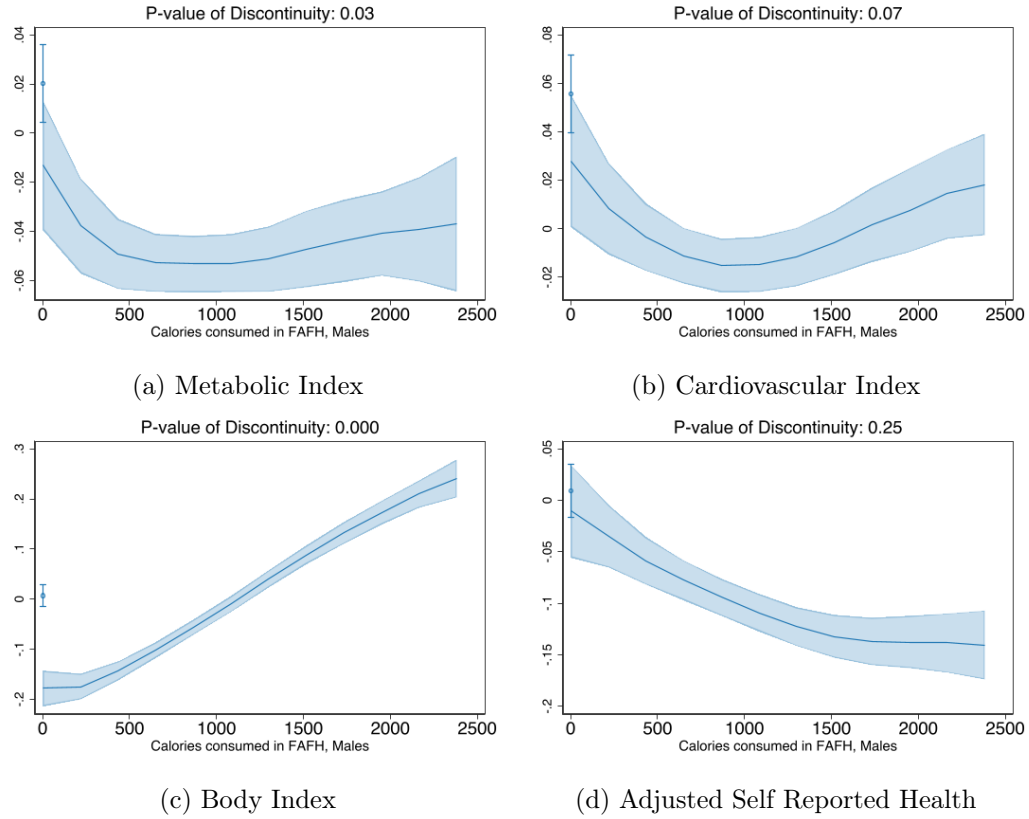


Figure B.2: Discontinuities at the bunching point for the male sample, using raw (unresidualized) outcomes against FAFH calories; the raw gaps are weaker than for females (cardiovascular $p = 0.07$, metabolic $p = 0.03$).

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Confidence intervals of 95% are shown on the plot, and the p-value of the discontinuity at $C = 0$ is shown on top of each panel.

The plot represents the effect of an additional 1,000 calories on each outcome variable, regressed on zero, and a bandwidth of 500 calories for other values.

Adjusted self-reported health is calculated by standardizing the self-reported health by wave.

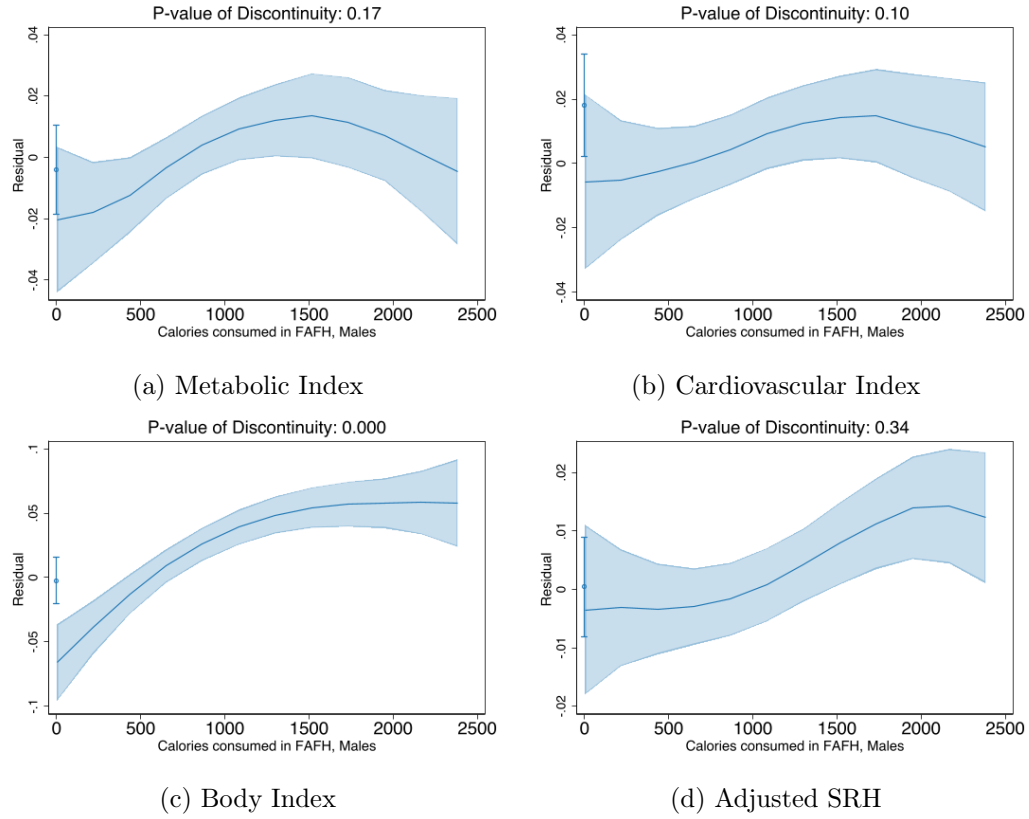


Figure B.3: Discontinuities at the bunching point for the male sample, using residualized outcomes (net of demographic controls) against FAFH calories: no gap remains significant at the 5% level once controls are added (cardiovascular $p = 0.10$, metabolic $p = 0.17$).

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The covariates on this model includes: age, age squared, poverty ratio, and dummies for education, wave, marital status, gender, and whether the individual is dieting or not, using a bandwidth of 500 calories. Confidence intervals of 95% are shown on the plot, and the p-value of the discontinuity at $C = 0$ is shown on top of each panel. Adjusted self-reported health is calculated by standardizing the self-reported health by wave.

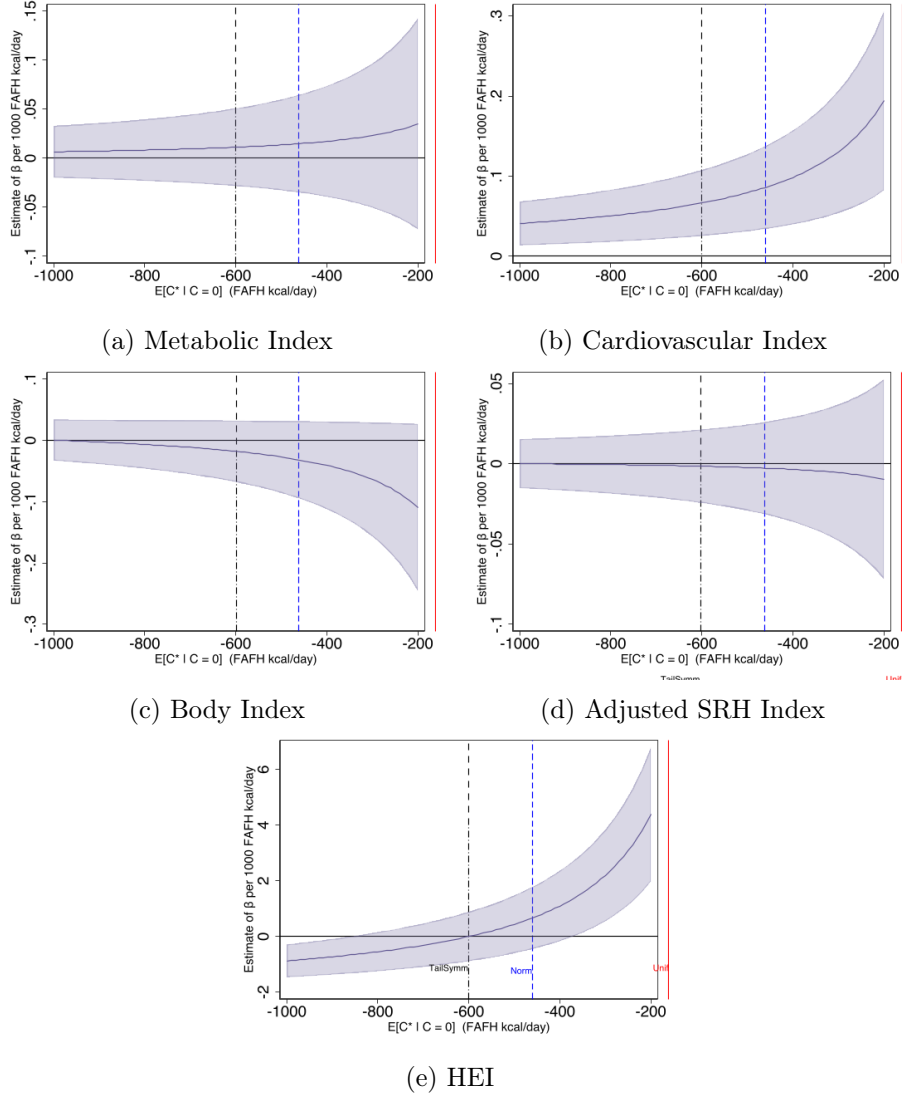


Figure B.4: Distribution-sensitivity check for the male sample: $\hat{\beta}$ for each outcome as the assumed counterfactual mean $\mathbb{E}[C^* | C = 0]$ is varied across the support implied by the three reference distributional assumptions.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: Each Figure shows the values of $\hat{\beta}$ for different values of $\mathbb{E}[C^* | C = 0]$, using the male sample. The vertical lines black, blue, and red represent the values for the expectation for semiparametric uniform, semiparametric normal, and nonparametric tail symmetric distributions, respectively.

Bootstrapped confidence intervals of 95% are shown on the plot, based on 100 iterations.

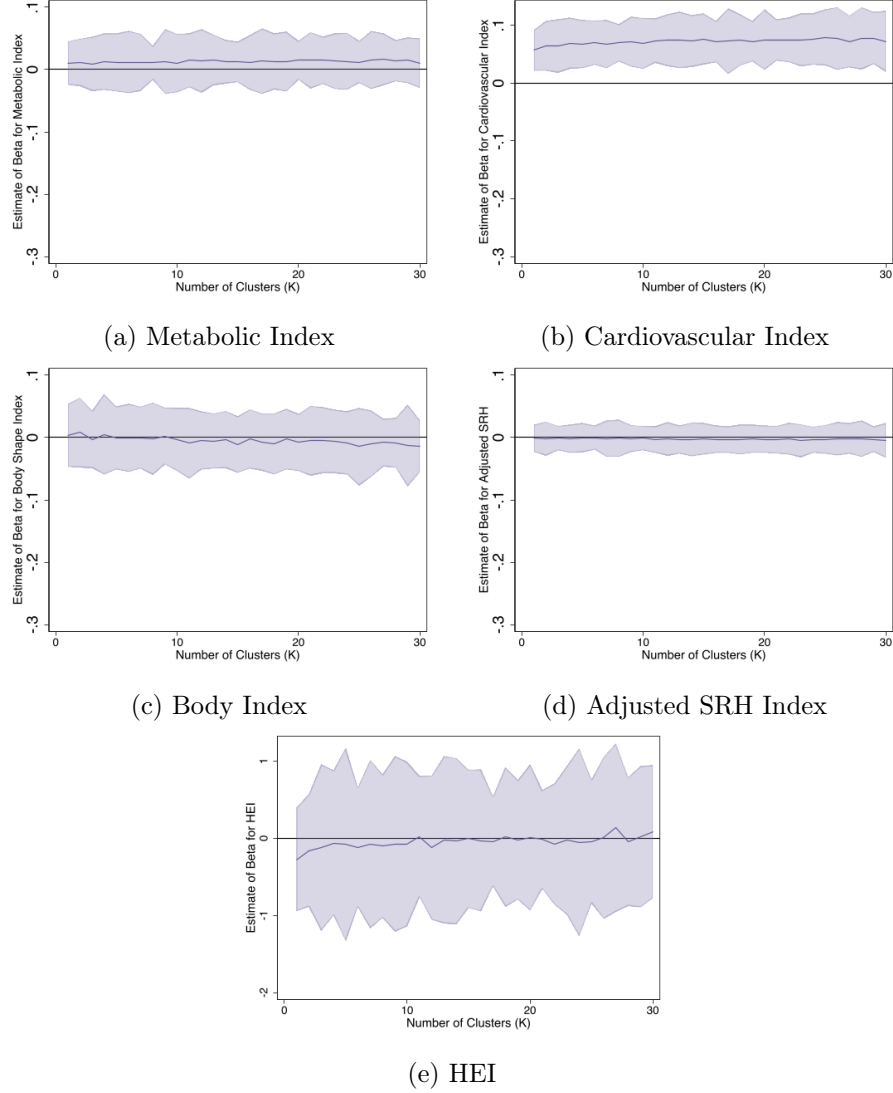


Figure B.5: Robustness check for the number of clusters K in the control function $m(X)$, for the male sample: $\hat{\beta}$ re-estimated for $K = 1, \dots, 30$.

Data source: National Health and Nutrition Examination Survey (NHANES)

Notes: The Figure shows the values of $\hat{\beta}$ for different numbers of clusters K , for the male sample. Bootstrapped confidence intervals of 95% are shown on the plot, based on 25 iterations.

Distributional assumption is the nonparametric tail symmetric distribution.